

Criminal Law, Investigation and Procedure Victoria
Investigation
Identification
Scientific identification of suspects
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[ID.3000] Introduction

Identification of a suspect can be carried out in a variety of ways. This section of the chapter deals with the powers of investigators to orchestrate the identification of a person suspected of criminal activity by means of fingerprinting, DNA profiling, intimate and non-intimate body sampling and physical examination.

Fingerprinting

[ID.3020] Fingerprinting – overview

In the 17th century anatomical studies of skin suggested identifying characteristics of fingerprints. This perception was consolidated in the mid-nineteenth century. By the turn of the century, the investigative value of fingerprinting was being recognised.¹ Dactyloscopy, or fingerprinting, forms one part of the broader science known as lophoscopy, the study of the development, classification and identification of the prints left by the papillary ridges of the skin.

The hairless parts of the body, the inner surfaces of the hands and the soles of the feet, are covered by patterns formed by raised ridges of skin, called papillary or friction ridges. Such patterns are formed by the foetus at an early stage of development and do not change during a lifetime. The prints of identical twins can be the same but otherwise fingerprints are uniquely identifying. The particular pattern exhibited by any given individual is partially genetically determined. Classification of fingerprints is done on the basis of arches, loops and whorls. Identification of

fingerprints relies on pattern matching and by the detection of particular ridge characteristics, also known as Galton details, points of identity or minutiae, and the comparison of the relative positions of such points and those found in a comparison sample. There are three ridge characteristics – the ridge ending, the bifurcation and the dot or island.

Latent fingerprints occur as a result of eccrine gland secretions. They are generally invisible to the naked eye but can become visible by the application of luminescence, powders and physico-chemical and chemical detection techniques.²

The Australian National Automated Fingerprint Identification System (NAFIS) is a database containing some 2 million fingerprints enabling ready search and comparison. Identification of a suspect varies by way of criteria, the numbers of identification points differing between a low end of 8, used in Victoria, and 17, used in Italy and France.

By 1913 fingerprinting as a scientifically sound means of establishing identity had achieved forensic acceptance:³

The leading facts respecting finger markings such as have been proved in this case may be regarded as established biological facts. The first of these facts is that finger markings remain absolutely unaltered from birth to death, however long a human being may live, unless, of course, destroyed by scars. The second is that there is no rule whatever of race or heredity which determines on what part of the finger print certain marks will be found which for convenience I will call “junctions” and “free ends”.

The definitive statement of the law in Australia is that of the New South Wales Court of Criminal Appeal in *R v Carr*:⁴

A fingerprint is as much a part of a person's physical characteristics as the nose on his face, a missing finger, a mole on his chin or a scar on his thumb. A witness may tell of these identifying characteristics. It is true that in the case of a fingerprint the whorls and loops on the skin will not normally be directly visible. They will need to be printed out on a sheet of paper and a magnifying glass will need to be used before they can be read. Nonetheless wherever a man goes he leaves fingerprints behind him, on his plate, on spoons, forks and cups, in fact whenever he touches a medium which is capable of absorbing them.

The science of fingerprint assessment has developed to the stage where properly trained and experienced technicians are regarded by the courts as able to recognise points of similarity which would not be at all clear to an uninstructed lay person:⁵

It is a matter for expertise not possessed by the ordinary run of mankind to identify characteristics of fingerprints and their patterns in each of two prints and make a comparison and form a conclusion as to whether they are identical or not and the jury could not be invited or allowed to act as experts.⁶

The leading modern case on fingerprinting is that of *R v Buisson*⁷ where a fingerprint identified on a block of hashish was vital evidence for the Crown. Evidence was given that identification could properly be done with as few as 7 or 8 points of identification, or even fewer in special circumstances. The witness said that 12 points had become accepted as a matter of practice in the New Zealand fingerprint section as the minimum to be proffered as indicative of identification of a suspect. The court noted that:

[I]n the experience of most – if not all – judges, it is the practice for police experts to put forward at least 12 points in common, and if a witness endeavoured to support identity on less he or she might well be asked for an explanation. However, there is no rule that an opinion based on a lesser number is inadmissible. No doubt a Judge would reject such evidence if the similarities on which it was based were so few or so unremarkable that they could have no probative value in determining identity.⁸

The court noted the following views expressed by Campbell:⁹

An expert, faced with a scene of crime mark which corresponds in every one of the required sequence of characteristics [16 in England] will say, without hesitation, that the mark was made by the person who supplied the known sample. Where there are less than the standard number, the expert will still hold a firm and reasonable belief that the marks are of common authorship until there comes a point where the identification becomes truly suspect. However, the only value such an incomplete identification can have is as a pointer to those concerned with investigating the crime. The danger in treating incomplete identifications as evidence is that certainty of fingerprint identification may be subject to wholly unjustified attack. What number of points in agreement justify a “probable” and how many are required to suggest “highly probable”? What magic raises high probability to certainty?

The Court of Appeal declined to stipulate that the jury should have been directed that they could find identity established only if they were satisfied that there were 12 points of identity or similarity. It referred with approval to the Manitoba case of *R v Nielsen*¹⁰ a footprint case, which permitted evidence to go to the jury that it was “highly probable” that the prints discovered at the scene were

identical to those of the defendant on the ground that “it no doubt added to the already cogent body of evidence pointing to” the defendant's guilt.

[ID.3040] Chief Commissioner's Instructions – fingerprinting

The *Victoria Police Manual – Operations* contain specific instructions for police in relation to fingerprinting. It is important to note that the Chief Commissioner's Instructions are just that – they are not the law although they are clearly intended to reflect the law in its practical application as far as possible. However, they do provide a sound basis for testing whether proper investigation procedures have been applied in a particular case. In particular the fingerprinting instructions provide useful information on the Victoria Police (VP) forms required for various aspects of the fingerprinting system and basic requirements for taking fingerprints.¹¹

[ID.3060] Power to fingerprint

At common law fingerprints could only be taken with the consent of the suspect; infraction often resulted in fingerprinting evidence being ruled inadmissible.¹² Police only possess those powers explicitly conferred upon them to take fingerprints.

Children aged under 10

Police cannot take fingerprints from children who are under the age of 10.¹³ In fact they cannot even request children of such an age to provide their fingerprints.¹⁴

Children aged 10 and under 15

Police can take fingerprints from a child aged between the ages of 10 and 15 where the child is reasonably believed to have committed, been charged with, or been summonsed to answer a charge for an indictable offence of a *Crimes Act 1958 Sch 7* summary offence. However, there is a further condition – the child or their parent or guardian must consent or when such consent is not provided, the Children's Court has to make an order permitting the taking of the fingerprints.¹⁵ A police member wishing to fingerprint a child between the ages of 10 and 15 must inform both the child and the parent or guardian of the child of a range of things, in language likely to be understood by each of them:

- the purpose for which the fingerprints are required;
- the offence which the child is believed to have committed or with which the child has been charged or for which the child has been summonsed to answer a charge;
- that the fingerprints may be used in evidence in court;
- that the child's parent or guardian may consent to the child's fingerprints being taken;
- that if such consent is not forthcoming an application may be made to the Children's Court for an order directing the child to give their fingerprints;

- that if the child is not charged with a relevant offence within six months or such a charge is not proceeded with or the child is found not guilty of the charge or any other relevant offence within that period, the fingerprints will be destroyed.¹⁶

The role of parents or those acting in the role of parents is important under the legislation. They must be present during the request for fingerprints, the giving of s 464L(3) information and during the taking of the fingerprints by consent.¹⁷ There are also obligations for tape-recording the provision of the s 464L(3) information if the child is in custody, as well as the responses of the child or the parent or guardian.¹⁸

The Children's Court may order a child of more than 10 but less than 15 to give his or her fingerprints if satisfied on the balance of probabilities that:

There are reasonable grounds to believe that the child has committed an indictable offence or a summary offence referred to in [Schedule 7](#); and in all the circumstances the making of the order is justified.¹⁹

[Schedule 7](#) refers to a very broad range of summary offences where the maximum penalty (whether for a first or subsequent offence) is or includes a period of imprisonment and in addition for:

- an offence under s 3(2) or (4) of the *Court Security Act 1980*;
- an offence under s 6(2) of the *Control of Weapons Act 1990*;
- an offence under s 97 of the *Police Regulation Act 1958*;
- an offence under s 36A of the *Drugs, Poisons and Controlled Substances Act 1981*; or
- an offence under s 20, 23 or 35(6) of the *Prevention of Cruelty to Animals Act 1986*.

The Children's Court is directed to take into account three matters in determining whether the making of an order that a child submit to the taking of his or her fingerprints is “justified”:

- (a) the seriousness of the circumstances surrounding the commission of the offence; and
- (b) the alleged degree of participation by the child in the commission of the offence; and
- (c) the age of the child.²⁰

However, a child the subject of a police application that the court grant an order for fingerprinting has limited rights of participation in the hearing. The child is not a party to the application, cannot call or cross-examine witnesses and, less importantly, is confined in terms of addressing the court to matters about which the court needs to be satisfied and the grounds upon which the court might conclude that the making of such an order is “justified”.²¹ Where the court determines that a child's fingerprints should be taken, it must give reasons for its decision²² and cause a formal record to be entered of its reasons.²³ Police may use reasonable force to take the fingerprints²⁴ and the court has power to issue a warrant authorising police to break and enter and search any

place where the child is suspected to be, to arrest the child and to take the child to the nearest accessible police station for fingerprinting.²⁵

Adults and children aged 15 or more

A member of the Victoria Police may take the fingerprints of a person 15 years or older if the suspect is reasonably believed to have committed, been charged with, or to have been summonsed to answer an indictable offence or a summary offence where the maximum penalty (whether for a first or subsequent offence) is or includes a period of imprisonment, for:

- an offence under s 3(2) or 3(4) of the *Court Security Act 1980*;
- an offence under s 6(2) of the *Control of Weapons Act 1990*;
- an offence under s 97 of the *Police Regulation Act 1958*;
- an offence under s 36A of the *Drugs, Poisons and Controlled Substances Act 1981*; or
- an offence under s 20, 23 or 35(6) of the *Prevention of Cruelty to Animals Act 1986*.²⁶

The criterion therefore is one of reasonable belief.

A member of the Victoria Police intending to fingerprint a suspect must inform the person of five matters, in language likely to be understood by them:

1. the purpose for which the fingerprints are required;
2. the offence which the person is believed to have committed or with which the person has been charged or for which the person has been summonsed to answer to a charge;
3. that the fingerprints may be used in evidence in court;
4. that if the person refuses to give his or her fingerprints voluntarily, a member of the police force may use reasonable force to obtain them; and
5. that if the person is not charged with a “relevant offence” within six months or is so charged but the charge is not proceeded with or the person is not found guilty of the offence or any other relevant offence before the end of that period, the fingerprints will be destroyed.²⁷

Section 464O(1) of the *Crimes Act 1958* defines a “relevant offence” extremely broadly to mean:

- (a) the offence in respect of which the fingerprints were taken; or
- (b) any other offence arising out of the same circumstances; or
- (c) *any other offence in respect of which the fingerprints have probative value* (emphasis added).²⁸

Reasonable force can be utilised to take fingerprints from a person from whom it is legitimate for them to be taken. However, this is only so if a person declines to give their fingerprints voluntarily. The use of reasonable force must be authorised by a member in charge of a police station at the

time of the request or a member of or above the rank of a sergeant, namely a senior sergeant, an inspector, senior inspector, superintendent etc.²⁹

[ID.3080] Taking, use and destruction of fingerprints

Taking of fingerprints

When fingerprints are taken pursuant to a court order or reasonable force is used to procure them, the legislation prescribes that, if practicable, they are to be taken by a police member of the same gender as the suspect.³⁰ In addition, it is prescribed that a member involved in investigating the offence for which the fingerprints are required must not, if practicable, take the fingerprints.³¹

Destruction of fingerprinting records

The Chief Commissioner is obliged by [s 464O of the *Crimes Act 1958*](#) to destroy fingerprinting records where:

- a person's fingerprints have been taken but they have not been charged for six months after that time;
- the person whose fingerprints have been taken has been charged but the charge has not been proceeded with for six months after that time;
- the person whose fingerprints have been taken has been found not guilty and a month has elapsed from the end of any appeal period.³²

However, a member of the police force can apply within the relevant time frames for one extension of the period for up to six months longer.³³ When records are destroyed, the suspect must be notified within 14 days.³⁴ Where the destruction as prescribed by [s 464O](#) does not take place or where unauthorised use of copying of fingerprints is done, it constitutes a summary offence punishable by a fine of up to 10 penalty units.

Where a child is fingerprinted, and the person is not found guilty before the age of 26, the Chief Commissioner must destroy the fingerprints.³⁵ A request must be made for destruction if the prints were taken before the commencement of [s 7 of the *Crimes \(Amendment\) Act 1993*](#). However, the obligation to destroy fingerprint records does not exist in relation to fingerprints “retained” in respect of a variety of serious physically and sexually violent offences.³⁶

Fingerprinting evidence

Fingerprinting evidence is prima facie inadmissible if the requirements of [ss 464K, 464, 464L, 464M and 464N of the *Crimes Act 1958*](#) are not complied with or if the prints or a record, copy

or photograph of them should have been destroyed but have not been.³⁷ However, a court has a residual discretion to admit such evidence if it is satisfied on the balance of probabilities that the circumstances are exceptional and justify the reception of the evidence or if the accused consents to the admission of the evidence.³⁸ The probative value of the fingerprints is not to be regarded as an exceptional circumstance.³⁹

[ID.3100] Fingerscanning for identification purposes

Where a person aged 15 years or over has been charged with an indictable offence or a relevant summary offence and is in a police station because of the charging or has been remanded in custody in relation to it a police officer may take a fingerscan only for the purposes of identification.⁴⁰ Fingerscans taken under s 464Q are not admissible in evidence in any proceeding.⁴¹

A member of the Victoria Police intending to fingerscan a person must inform the person, in language likely to be understood by them, that the fingerscan is:

1. only for the purpose of identifying them; and
2. inadmissible in evidence.⁴²

If the person refuses to give his or her fingerscans voluntarily, a member of the police force may use reasonable force to obtain them, after informing the person that reasonable force may be used.⁴³ A person of the same sex as the person to be fingerscanned must take the fingerscan if practicable.⁴⁴ Such fingerscans need not be destroyed but may be retained on a database for use by persons authorised in writing by the Chief Commissioner.⁴⁵

DNA

[ID.3120] What is DNA – DNA profiling, nuDNA, mtDNA, Y Chromosome DNA, trace DNA and LCN typing

DNA or deoxyribonucleic acid is where the blueprint for life is stored.⁴⁶ It is essentially the same in all cells of the human organism, consisting of two complementary strands of material wound around each other in a double helix. It is made up of deoxyribose, phosphate and four bases, adenosine (A), thymine (T), guanine (G) and cytosine (C). These four bases behave in a certain way – so that A always pairs up with T and G with C. There are two main types of DNA that are of interest in forensic science – Nuclear DNA (nuDNA, which comprises autosomal, X chromosome and Y chromosome DNA) and Mitochondrial DNA (mtDNA). Identification in criminal investigations is predominantly concerned with autosomal and Mitochondrial (mtDNA) DNA. However, Y chromosome DNA is potentially useful in some limited circumstances in the context of a criminal investigation. At the present time there is no significant possibility

of X chromosome DNA having a role in criminal investigations.⁴⁷ There are approximately three billion base pairs on the DNA strand and it is their sequence which is responsible for a person's characteristics. About three million sites vary from one person to another. In humans the DNA is packaged into 22 pairs of chromosomes found in both males and females, the autosomal chromosomes, plus the sex chromosomes, the XX in females and the XY in males.

Certain properties of DNA make it a very useful tool in criminal investigation:

- with the exception of identical (monozygotic) twins each individual's nuDNA is unique;
- with the exception of gametes (spermatozoa and ova) the DNA in each cell of an individual is identical to that in every other cell and remains constant throughout their life (unless a mutation occurs);⁴⁸
- DNA is very stable and may remain unaltered for many years even in adverse environments;
- DNA is found in virtually all human tissue or bodily secretions;
- DNA analysis is possible even from very small samples containing only relatively small numbers of cells.

It is to the sequencing in the DNA that testing has regard in searching for matches in samples. An early form of DNA testing, introduced in the 1980s, is via restriction fragment length polymorphism or RFLP technology. It receives its name by its use of restriction enzymes which cut DNA into fragments of a suitable size for analysis. Site-specific variation could be discerned by using a short piece of DNA as a probe. Probes consist of approximately 20 sugar-phosphate moieties, each with a specific base. They are specific for a particular allele, a sequence of DNA.

A technique which began being used in Victoria in the second half of the 1990s for forensic purposes enables the multiplication of extremely small amounts of crime scene sample so as to enable testing. This has also reduced a number of the problems which had previously existed with DNA profiling. The technique is known as polymerase chain reaction or PCR.

Mullighan J in *R v Jarrett* described the PCR technique thus:⁴⁹

The polymerase chain technique (PCR) is a technique which amplifies or copies a small region of DNA millions of times. Only a small number of the bases are looked at by the PCR method. By using this method the type of DNA chosen is very variable from person to person. Once the chosen region is copied up there is produced DNA fragments from each strand, one from each parent. One is longer than the other and they are separated. The PCR method provides a way of measuring the length of these two fragments which are known as alleles and their size is described by numbers. The two alleles together provide the genotype of a person.

Gutowski explains that PCR:⁵⁰

[M]imics natural DNA application but concentrates on the speed of replication of a short DNA region. Double-stranded DNA is first denatured to its component single strands. Conditions are altered from denaturing to those that favour re-annealing (hybridisation). Short oligonucleotide primers are added. These primers then anneal to the component single strands to give partially double-stranded DNA. Conditions are then changed again to favour the extension of the DNA primers and the double-stranded areas are then extended. DNA sequence extension is directional and the primers are therefore constructed so that one primer is elongated along one strand in one direction and the other primer is elongated along one strand in the opposite direction. After an appropriate period, elongation is stopped, conditions changed again to denature the DNA and the whole cycle repeated, time after time. The end result can easily be a one millionfold amplification of a small segment of DNA in a few hours.

RFLP typing is now little used. DNA profiling now generally depends upon PCR techniques. Much of the variation exploited by the PCR techniques, however, still uses length variation depending on the presence of different numbers of tandem repeat DNA sequences, or Short Tandem Repeats (STRs), in different individuals.

Since 1998/1999 the Profiler Plus system has been used by all State forensic science laboratories in Australia. Mullighan J has said of the system that:⁵¹

It uses the same PCR, fluorescence and electrophoresis technology with one of the instruments and Genescan, as well as additional software known as Genotyper. The system includes kits provided by Perkin Elmer [the distributor] which include all reagents needed for amplification, primers which are specific for the various loci, PCR reaction buffer and particular DNA polymerase as well as the requisite allelic ladders ... The amplified fragments of the DNA are separated by electrophoresis and then detected by laser and recorded by camera and collected by computer software, as is the case with the earlier multiplex and megaplex systems. The Genescan software interprets the collected data and displays it ... which is interpreted by analysts. The Genotyper software designates the alleles at each locus and ascribes to them a size and an RFU level which is displayed in a graphic presentation in the form of an electropherogram. There are some differences between Profiler Plus and other systems. The main differences ... are that some of the loci are different from those used in any one of the other systems and the use of Genotyper. In my view, it is inaccurate to describe Profiler Plus as a totally new system of DNA analysis in the same way that such a description could have been given to STR technology and fluorescence technology when they emerged. The Profiler Plus system involved the processes familiar to forensic scientists of extraction, quantification, electrophoresis and detection by automated sequencers and assistance in interpretation by computer software.

It is important to have regard to when DNA testing was done in order to appreciate the technique likely to have been employed. Considerable change has taken place in testing methods and laboratory protocols in response to serial critiques and controversies which have surrounded the forensic application of DNA technology.⁵² Advances in technology have led to improvements in performance of DNA analysis in criminal casework, and this is an ongoing process.

[ID.3140] Types of DNA – autosomal DNA, mitochondrial DNA and Y chromosome DNA

In higher organisms, including humans, every cell contains two types of DNA – that which is found in the nucleus of the cell (nuclear DNA or nuDNA) and that which is found outside the nucleus in the mitochondrion (mitochondrial DNA or mtDNA). A nucleus contains forty-six chromosomes, bearing two copies of a person's nuclear DNA. There are twenty-two pairs of chromosomes called autosomes – one of each pair being inherited from the mother and one from the father – and the sex chromosomes – two X chromosomes in females and an X and a Y chromosome in males. With the exception of identical twins, nuclear DNA is unique to an individual (hence nuDNA is referred to as a “unique identifier”). On the other hand there may be as many as 1000 mitochondria in a cell with several copies of mtDNA in each mitochondrion. Consequently there is a much greater abundance of mtDNA in a cell than there is nuDNA. While nuDNA is inherited from both parents, mtDNA is inherited only from the mother. As a result all those in the maternal line – maternal siblings, children of daughters, maternal aunts or uncles and maternal cousins born to maternal aunts all have the same mtDNA (hence mtDNA cannot be said to be a unique identifier).

In some situations, such as where the samples are small, have a low amount of DNA or have degraded, the vast number of mitochondria in a cell compared to a single nucleus means the use of mtDNA has a significant advantage. It has been said that mtDNA is not, and never will be, more discriminating than nuclear DNA.⁵³

Y chromosome DNA is found only on the male sex (Y) chromosome. It is passed from father to son through the paternal bloodline and so is present only in males. All males in the paternal line – paternal grandfather, father, sons, father's brothers – have identical Y chromosome DNA. Consequently Y chromosome DNA testing can be very useful in some limited circumstances. This may occur for samples where a high level of female DNA is mixed with a much lower amount of male DNA. Testing for autosomal DNA may result in the female DNA masking the presence of any other profile. Potentially testing for Y chromosome DNA could reveal any male DNA present whilst the female nuDNA would not be detected. The result cannot be a discriminating as autosomal DNA profile, as similar to mtDNA, the Y chromosome DNA profile is not a unique identifier. It can only identify the sample came from a male in a particular paternal line. Nevertheless, Y chromosome DNA may provide valuable information for investigators and potentially could be significant evidence given other evidence in a case.

While in general terms the methodology used in both nuclear DNA and mtDNA analysis is the same – extraction, analysis, comparison and statistical analysis – there are significant limitations to mtDNA analysis compared to nuclear DNA analysis. In Australia at present autosomal DNA testing enables analysis using either a 10 marker profiling kit known as Profiler Plus (“P+” which uses of nine DNA sites or loci plus the gender site) or, since the beginning of 2013, a much more sensitive and discriminating 21 marker profiling kit known as PowerPlex21 (“PP21” which uses 20 DNA sites or loci plus the gender site). In comparison whereas mtDNA analysis involves only two sites (hypervariable regions 1 and 2 of the control target region in mtDNA). The statistical analysis used in relation to autosomal DNA involves the use of frequency calculations based on competing hypotheses (broadly speaking the prosecution hypothesis and the anticipated defence hypothesis). The analysis involves comparing the likelihood of the specific DNA match with DNA from an identified person compared with a match with DNA from a person of the same sex chosen at random. The result provides an estimate applicable to the whole population. In March 2013 a more sophisticated statistical software tool, STRmix, was introduced by the Victorian Police Forensic Science Service (VPFSS) to account for the random (stochastic) effects in autosomal DNA analysis including “drop in” and “drop out” of alleles in a DNA profile. The STRmix tool is described as “fully-continuous” in that it makes use of all information in a DNA profile, including peaks in the profile that other statistical tools would exclude because they fall below a designated stochastic threshold.⁵⁴ Statistical analysis used in relation to mtDNA involves presenting the number of times the mtDNA type has been seen in the selected database (international or localised) given it has been seen in the reference sample. In mtDNA analysis the mtDNA types are non-random because they are maternally inherited and mtDNA types in families are not randomly distributed in the population. It is therefore problematic to extrapolate from the frequency in the database to the frequency in the whole population. As a consequence, all other things being equal, the “power” of the statistical analysis achieved by mtDNA is less than that achieved by nuDNA analysis. Of course in the real world all other things are not equal and the significance of a statistical analysis based on any DNA comparison must be determined from all the circumstances of a particular case. Statistical analysis of Y chromosome DNA is done in a similar way to analysis of mtDNA results. Currently the test kit for Y chromosome DNA identifies seventeen loci on the Y chromosome.⁵⁵

[ID.3160] Trace DNA – Low copy number typing

The term Trace DNA is generally used to mean DNA the origin of which may be invisible or unknown (in contrast to DNA from blood, semen or other identifiable tissue). Improved technology has enabled successful typing of DNA to be pushed to ever smaller amounts of DNA, even down to the level of typing DNA contained in a single cell. Typing from extremely limited DNA samples is sometimes called low copy number (LCN), high sensitivity or low template DNA (LTDNA) typing. For STR analysis the amount of DNA required is typically in the range of 200 picograms (pg) to 2 nanograms (ng).⁵⁶ LCN typing typically involves samples of below 100–200 pg of DNA.

However, the specific procedures that have been devised for the interpretation of LCN or LTDNA profiles have wider application in the evaluation of any low-level DNA profile.

Whilst new technology makes it possible to type minute quantities of DNA that are invisible to the human eye, these technologies raise significant issues for the criminal justice system. LCN typing inherently provides non-reproducible results. Reproducibility is the cornerstone of reliability for science. The inability to reproduce results in LCN typing is a complex matter beyond the scope of this discussion. In short it arises partly from stochastic effects (random variability) in the PCR phase of the analysis, resulting in inconsistent amplifications of certain alleles.⁵⁷ This manifests as a greater potential for error in interpretation and the consequent need for caution in making resultant statistical probability estimates. General acceptance of LCN methodology, along with reliability, is one of the requirements for opinion evidence based on specialist knowledge, but general acceptance does not make scientific method valid and reliable – the converse is also true. The gold standard for DNA analysis is based on the validity and reliability testing of DNA methodology.⁵⁸

Locard's principle, with every contact between two items there will be an exchange (trace evidence), applies to reference sample collection and testing procedures as much as it does to crime scene examination and criminal activity.⁵⁹ The risk of contamination is, at least notionally, significant when trace DNA and LCN typing is involved. Given the minute quantities of DNA material involved in LCN typing great care must be taken to avoid inadvertent contamination at all stages in the process from crime scene isolation, examination and sample collection to laboratory testing. Generally speaking it is other evidence, or the lack of it, in an investigation that is likely to highlight the likelihood, or otherwise, of contamination being an issue.

Nevertheless, LCN typing is a valuable tool for forensic scientists. In many instances LCN typing is used to assist investigators identify potential leads and human remains without making its way into evidence in court.

The relevant protocols for collecting, storing and testing DNA samples and related items are not publicly available – see discussion of this issue at [\[ID.3240\]](#).

[ID.3180] Development of Australian case law – when opinion is admissible as expert testimony

It can now be generally accepted in Australia that DNA evidence is *prima facie* admissible if properly qualified as an expert body of evidence.⁶⁰ As noted it is admissible as an exception to the opinion rule as an opinion based on specialised knowledge.⁶¹ In order for expert testimony to be admissible:

- the subject matter of the opinion must be relevant to a fact in issue;

- the subject matter of the opinion must be such that an ordinary unqualified person would not be able to form a sound judgment without the assistance of a witness possessing special knowledge or experience in the area;
- the subject matter of the opinion must form part of a body of knowledge or experience sufficiently organised or recognised so as to be accepted as a reliable body of knowledge or experience; and
- the witness must have acquired by study or experience sufficient knowledge in the subject matter to make his or her opinion of value in resolving the issues before the court;⁶² and
- the witness's opinion must be wholly or substantially based on that knowledge.

The factual foundation for the expert opinion must be proved by admissible evidence or the assumptions as to fact on which it is based stated explicitly.

The High Court has observed:⁶³

The admissibility of opinion evidence is to be determined by application of the requirements of the *Evidence Act 1995 (NSW)* rather than by any attempt to parse and analyse particular statements in decided cases divorced from the context in which those statements were made. Accepting that to be so, it remains useful to record that it is ordinarily the case, as Heydon JA said in *Makita (Aust) Pty Ltd v Sprowles* (2001) 52 NSWLR 705; 25 NSWCCR 218; [2001] NSWCA 305 at [85], that “the expert’s evidence must explain how the field of ‘specialised knowledge’ in which the witness is expert by reason of ‘training, study or experience’, and on which the opinion is ‘wholly or substantially based’, applies to the facts assumed or observed so as to produce the opinion propounded”.

To be relevant the opinion, if accepted, must be capable of rationally affecting, directly or indirectly, the assessment of the probability of the existence of a fact in issue in the proceeding.⁶⁴

The point at which a scientific principle moves from being experimental to the basis upon which a court will act is when it is “sufficiently established to have gained general acceptance in the particular field in which it belongs” (the Frye test).⁶⁵ Where an expert witness applies new or unfamiliar technology or techniques, the court may need to be satisfied the results produced by the technology or technique are properly considered part of that field of expert evidence.⁶⁶

Australian courts have not adopted the approach of the United States Supreme Court which requires an assessment of evidentiary reliability as part of the determination of whether proposed scientific evidence is based on knowledge:⁶⁷

... in order to qualify as “scientific knowledge”, an inference or assertion must be derived by the scientific method. Proposed testimony must be supported by appropriate validation

– ie, “good grounds”, based on what is known. In short, the requirement that an expert’s testimony pertain to “scientific knowledge” establishes a standard of evidentiary reliability.

.... Faced with a proffer of expert scientific testimony, then, the trial judge must determine at the outset, pursuant to Rule 104(a), whether the expert is proposing to testify to (1) scientific knowledge that (2) will assist the trier of fact to understand or determine a fact in issue. This entails a preliminary assessment of whether the reasoning or methodology underlying the testimony is scientifically valid and of whether that reasoning or methodology properly can be applied to the facts in issue

This approach was expressly rejected by Spigelman J, when, observing he did not intend to suggest the United States cases had anything useful to say about [s 79 of the Evidence Act 2008](#) he concluded the focus of attention must be on the words “specialised knowledge” and not on the introduction of an extraneous idea such as “reliability”.⁶⁸

The approach of the United States Supreme Court requires consideration of validation studies, peer review, error rates and the existence and maintenance of standards as well as whether there is an expressed degree of acceptance within the relevant scientific community when assessing whether the evidence is based on specialised knowledge and admissible.⁶⁹ The High Court’s reasons in [Honeysett v The Queen \(2014\) 311 ALR 320; \(2014\) 88 ALJR 786; \[2014\] HCA 29](#),⁷⁰ an identification case based on anatomical comparison, reinforce the rigour required when assessing:

- what fact in issue the opinion evidence is said to prove, or assist proof of;
- whether the person’s training, study or experience has resulted in the acquisition of relevant knowledge; and
- whether the opinion expressed is substantially based on the witness’s specialised knowledge based on the person’s training, study or experience rather than observations and knowledge of everyday affairs and events.

In [Honeysett \(2014\) 311 ALR 320; \(2014\) 88 ALJR 786; \[2014\] HCA 29](#) the High Court concluded the evidence was not based on specialist knowledge. It found the case was not an appropriate vehicle for examining the larger issue whether there was a requirement for an independent means of validation before an opinion can be said to be based on specialised knowledge and therefore did not address the need for courts to assess evidentiary reliability as part of the determination of whether proposed scientific evidence is based on knowledge.⁷¹

Based on the Expert’s Code of Conduct contained in the [Uniform Civil Procedure Rules 2005 \(NSW\)](#) it has been suggested:⁷²

In thinking about compliance with [s 79\(1\)](#) and the Codes of Conduct, prosecutors and judges should consider whether non-compliant reports and testimony actually enable them

to answer the questions advanced by the High Court in *Dasreef*. What is the specialised knowledge, is the opinions based on it, and is the specialised knowledge linked to the individual's training, study or experience.

See discussion at [\[ID.2500\]](#) and [\[EA.PT3.3.20\]](#) ff on opinion evidence.

[ID.3200] DNA profiling evidence: admissibility

Since the early 1990s DNA profiling evidence has become an integral part of many criminal prosecutions. It has also enabled a number of people to have their convictions quashed⁷³ and unknown numbers of people excluded as possible suspects at the investigation stage.

The difficulties presented to legal practitioners and courts by DNA evidence should not be underestimated. The complexity and foreign nature of scientific and statistical processes associated with DNA evidence will cause most practitioners problems when assessing DNA evidence. The need for a thorough, rigorous and disciplined approach is made amply clear from an examination of the reasons of the Australian Capital Territory Court of Appeal in the re-hearing of the appeal in *Hillier v The Queen*.⁷⁴

Recognition of the challenges DNA evidence presents for lawyers, judges and juries led the Australian Capital Territory Court of Appeal to comment:⁷⁵

This case indicates a need for the urgent preparation of universally acceptable protocols so as to ensure, among other things, that there is full and unmistakably explicit disclosure to the defence of all DNA testing steps and results, and pre-trial exchange of expert reports about DNA evidence.

The need for rigor by all those involved in the collection, transporting, storing, laboratory testing, in court presentation and testing of DNA evidence, and the burden such evidence can place on judicial officers, is apparent from the report on the inquiry into the Victorian case of *Jama*⁷⁶ discussed below. It is critical the proper role of forensic science within the criminal justice system is understood by all the participants and all concerned conduct a critical examination of its reliability and probative value to a fact in issue. The specialised knowledge employed by forensic witnesses places a primary obligation on them:⁷⁷

Accordingly, in my view, it is not properly open to a scientist, on a later occasion when [their] trial evidence is being reassessed, to say, for example; "I answered correctly such questions as I was asked. If I had been asked other and more relevant questions I would also have answered them correctly. It is not my fault that the correct questions were not asked of me".

It is also critical prosecution and defence lawyers access protocols and other documentary evidence governing collecting, transporting, storage and testing of DNA samples relevant to the facts in issue of the case at hand. See discussion at [ID.3240] “Contamination and admissibility”.

Admissibility – general rule

As a general rule it can now be accepted that DNA evidence is *prima facie* admissible if properly qualified as an expert body of evidence. It is admissible as an exception to the opinion rule as an opinion based on specialised knowledge.⁷⁸ In this context DNA evidence means comparing the profile of DNA extracted from a sample collected at a crime scene (or related site) with the profile of DNA extracted from a relevant reference sample (usually from the accused or an alleged victim) coupled with a statistical interpretation of the extent of the match between the crime scene sample and the reference sample.

It is a matter for the jury or other tribunal of fact to evaluate any challenge to the evidence, including the views of competing expert witnesses, to determine whether they accept the evidence and, if so, what weight to attribute to it in all of the circumstances of the instant case.

In my view, admissible expert testimony relevant to a fact in issue and of probative value may not be withheld from the jury. Once the expert testimony is found to be admissible, it should be admitted even though challenged and it is a matter for the jury, properly and adequately directed, to decide in the context of all of the evidence in the case, whether the evidence should be accepted.⁷⁹

A substantial jurisprudence has evolved internationally in relation to the admissibility of DNA profiling evidence. A series of decisions in Australia and in England,⁸⁰ Canada,⁸¹ New Zealand⁸² and the United States⁸³ has established the admissibility of DNA profiling evidence provided that it is made amenable to evaluation by the triers of fact.⁸⁴ Hunt CJ at CL and Hidden J, for instance, in *R v Pantoja* noted that:⁸⁵

DNA testing has been accepted by the courts for some years as an acceptable scientific technique for the identification of the source of bodily tissues...

In order to be admissible it must be relevant to a fact in issue and comply with the well-established pre-conditions for expert evidence. Relevance to a fact in issue must be the starting point for any assessment of the admissibility of DNA evidence. There are many cases where the relevance test has not been satisfied.⁸⁶

It may be necessary, where the DNA evidence is challenged, for the prosecution to call the scientists or laboratory technicians who carried out the laboratory procedures. Evidence from another expert they gave instructions for the work to be done, does not overcome the hearsay rule. The factual foundation for the expert opinion must be proved by admissible evidence or the assumptions as to fact on which it is based must be stated explicitly.⁸⁷ Given the importance of the DNA evidence to the prosecution case and its compelling nature:⁸⁸

I do not think the matter of the correct carrying out of testing procedures should normally be proved, over objection, merely by evidence of the existence of the procedures and the giving of instructions, and otherwise left to inference. If for any reason the persons who actually did the work are unavailable, there may be justification for such a course. But there is no suggestion of that here.

The development of the Australian case law, including the question of what is required for expert testimony to be admissible, and examples of situations where DNA evidence was not admitted are discussed below.⁸⁹ For commentary on admissibility of evidence based on specialised knowledge see [EA.79.40]–[EA.80.20] and Odgers S, *Uniform Evidence Law* (Thomson Reuters, subscription service)

Part 3.11 “Discretionary and Mandatory Exclusions” of the *Evidence Act 2008* applies to DNA evidence.

[ID.3220] Development of Australian case law – challenging admissibility

Relevance is the starting point for any consideration of admissibility.⁹⁰ Once the requirement of relevance is satisfied issues of discretionary or mandatory exclusion may arise pursuant to [Part 3.11 of the *Evidence Act 2008*](#).⁹¹

The court has a legitimate concern that evidence is capable of being understood by a jury of ordinary citizens. There will be “unusual cases in which the judge has reason to fear that the jury will be overawed by the scientific garb in which the evidence is presented and will attach greater weight to it than it is capable of bearing”.⁹²

Expert DNA evidence must be excluded where the danger of its unfair prejudicial effect outweighs its probative value.⁹³ Many of the cases discussed here were decided before the UEA and are based on the common law. Nevertheless, they provide valuable guidance to the application of [s 137 of the *Evidence Act 2008*](#).

Mullighan J of the South Australian Supreme Court in *R v Jarrett* held that:⁹⁴

[I]t is no function of the Trial Judge, in advance of the trial, to assess expert witnesses and testimony and withhold evidence from the jury if of the view that the jury could not resolve the conflict or understand the evidence.

Nevertheless, it should be noted Mullighan J was referring to admissible expert testimony relevant to a fact in issue and of probative value.⁹⁵ He allowed that expert evidence may be excluded in the exercise of the courts discretion such as where its prejudicial effect outweighs its probative value, albeit he found it difficult to envisage circumstances where admissible expert evidence bearing on an important fact in issue could have a prejudicial effect greater than its probative value.⁹⁶

It is highly unlikely that a trial judge will exclude expert evidence about the interpretation to be given to DNA profiling unless expert evidence is adduced by the defence to call into question the evidence adduced by the Crown.⁹⁷

In a number of cases, both prior to and since Mullighan J's ruling in *R v Jarrett*, expert DNA evidence has been held to be inadmissible because the prejudicial effect of the evidence was found to outweigh its probative value.

Should the DNA evidence be so confusing or misleading due to the way in which the issues in dispute have been presented to the jury, a judge may find the expert evidence inadmissible.⁹⁸ In *R v Lucas*,⁹⁹ Hampel J observed that although the conflicting opinions of the expert witnesses were normally matters of weight he found it difficult to see how, in the circumstances of the case before him, a jury could resolve the different expert views placed before it and, if it were, whether it would be able to relate the differences meaningfully to any conclusion about the reliability of the DNA testing and the significance of the results.¹⁰⁰ Hampel J ruled the evidence of the DNA testing, its results and the conclusions drawn from it inadmissible because it lacked sufficient probative value compared to its possible prejudicial effect. Ultimately Hampel J reached this conclusion because there was no means for the jury to evaluate the evidence in the absence of evidence of the frequency of the match in the general population.¹⁰¹ However, it will be rare for expert DNA evidence to be ruled inadmissible and it might be argued that in *R v Lucas* the evidence was incomplete and thus not properly described as expert DNA evidence. As Hampel J noted, the more authoritative approach is that the weight to be given to different opinions about the interpretation to be accorded to the results of DNA profiling tests is a matter for the triers of fact and not for an admissibility determination. Considered in this light, it might therefore be argued there is no significant difference in the approaches of Mullighan J and Hampel J.¹⁰²

Bleby J in *R v Humphrey*¹⁰³ adopted the comments of Mullighan J in *R v Jarrett* that the resolution of conflicting expert testimony was a matter for the jury.¹⁰⁴ However, Bleby J went on to admit the evidence of the frequency of the relevant DNA profile occurring in the general population provided a foundation was laid as to the validity of the databases used to make the calculation. He noted that if the justification for the opinion is not led, where it is in dispute, the probative value of the evidence may be substantially outweighed by its prejudicial effect and call into play the discretion to exclude it.¹⁰⁵ This is consistent with Hampel J's approach, although he did not refer to *R v Lucas*.

In *R v Juric* [2003] VSC 382 Nettle J observed:¹⁰⁶

There remains the question of whether the statistical evidence should be excluded on the basis that a probability of one in twenty or one in seven is statistically insignificant, but of considerable prejudicial effect. As the decision of the Court of Appeal in this case implies, the difficulties which attach to potentially misleading or confusing evidence should ordinarily be dealt with by the use of appropriate directions. Nevertheless, there must and does come a point at which some technical evidence has the capacity to be so misleading or confusing, whatever directions may be given about it, that its probative value is exceeded by its prejudicial effect. At that point it is to be excluded from the jury.

In my opinion, one reaches that point in this case with the statistical calculation that it is twenty times more likely that the DNA came from the deceased and the accused than that it came from the deceased and some one else drawn from the Caucasian population of Victoria. That seems to me to be evidence that is so inherently capable of misleading or confusing the jury, whatever directions may be given, that it should be excluded.

When the evidence is such that a jury cannot evaluate the strength of the evidence by reference to its factual or scientific basis then its prejudicial effect is very likely to outweigh its probative value. While it is for the jury to evaluate conflicting expert opinions, where there is no basis upon which the jury can evaluate the competing opinions it is a matter for the judge to determine whether the evidence is admissible. In *R v Juric*¹⁰⁷ the evidence before the court involved two separate tests conducted two years apart on the same crime scene exhibit, a glove tip, using two different testing systems. The results from the tests using the Profiler Plus system were such that the results were open to competing hypotheses none of which could be preferred to another on any scientific basis.

The evidence will not necessarily become inadmissible simply because experts express differing opinions upon the factual or scientific bases in respect of which their opinions are expressed. However, the admissibility of such evidence must depend upon the judge's satisfaction that the jury can, on the basis of material put before them, properly and

reasonably evaluate the differing opinions expressed and make a responsible determination as to which of them is to be preferred.¹⁰⁸

Where there are clearly competing hypotheses for the presence of the DNA, one of which is consistent with innocence, then it would be necessary for the Crown to call evidence capable of laying a scientific foundation for the conclusion the presence of the DNA was explicable on the inculpatory rather than the exculpatory basis. In *R v Joyce*¹⁰⁹ evidence of the defendant's DNA on the complainant's clothing was capable of explanation by transfer in the normal course of activities within the defendant's home or by contamination at the time police collected the clothing from the complainant. Angel J, agreeing with *R v Juric*, held the DNA evidence had no probative value and was therefore inadmissible and that, in any event, its prejudicial effect would outweigh any probative value even if admissible.¹¹⁰

In *Director of Public Prosecutions (Vic) v Wise* [2016] VSCA 173¹¹¹ the court observed:

Moreover, one of the dangers associated with DNA evidence, is what has come to be known as the “CSI effect”. The “CSI effect” is a reference to the atmosphere of scientific confidence evoked in the imagination of the average juror by descriptions of DNA findings. As we have explained, as a matter of pure logic, the DNA evidence has little or no probative value. By virtue of its scientific pedigree, however, a jury will likely regard it as being cloaked in an unwarranted mantle of legitimacy – no matter the directions of a trial judge – and give it weight that it simply does not deserve. The danger of unfair prejudice is thus marked, and any legitimate probative value is, at best, small.¹¹²

An expert witness must be able to express their opinions in a comprehensible manner that is able to be understood by the trier of fact. In *R v McNeill (No 2)*¹¹³ the expert witness's reports were all but incomprehensible and in viva voce evidence the witness expressed herself in a manner that a jury would have difficulty understanding. There was a serious danger some members of the jury might give the evidence a particular meaning it was not intended to have and be misled into thinking the evidence actually went beyond what it did. Held: the danger of permitting evidence to be given in that form outweighed its potential probative value and the evidence was excluded.¹¹⁴

The significance of a match between the DNA at one or more loci in the crime scene sample with the same loci in the reference sample is given probative value by the statistical analysis and has been referred to as the “pivotal element” of DNA analysis.¹¹⁵ Where there is no statistical analysis to put before the jury it may mean, in the circumstances of a particular case, the probative value of the DNA evidence cannot be evaluated by the jury and its probative value is outweighed by its unfair prejudicial effect. However, it does not necessarily follow this is the case and there may be

other reasons why the evidence should be admitted, for example to establish that the victim of an alleged murder could not be the source of a relevant hair.¹¹⁶

Where the DNA evidence from a swab of the victim's clothing excluded as a contributor a prosecution witness who alleged his friend (the accused) had confessed to him shortly after the murder he had killed a person in the same location, but did not exclude the accused as a contributor, the evidence was excluded. No specific statistical analysis was given but the expert witness conceded there would be other people who similarly couldn't be excluded. The same witness conceded the "handy" DNA (ie skin cell or skin DNA transferred by a person touching the clothing) could have come from someone coming in contact with the victim at a hotel prior to his death. In ruling the evidence inadmissible Mildren J noted the prosecution had conceded it relied on an inference the crime scene DNA sample was left by the perpetrator. Mildren J found there was little or no evidence to support that inference and the DNA evidence was inherently weak and tenuous. Held: the evidence was clearly prejudicial, of no probative effect and there was an unacceptable danger the jury might use the evidence improperly notwithstanding careful directions. The whole of the DNA evidence was excluded.¹¹⁷

[ID.3240] Contamination and admissibility

The word "contamination" in this context is used to denote the presence of any DNA in a sample that has been introduced post-incident. Contamination of a crime scene sample or reference sample can be characterised in different ways. It may be sinister, a natural occurrence or reflect human error. It may arise at the crime scene during or after the alleged offence, it may arise during the investigation phase prior to, at the time of or after collection or it may arise during the laboratory phase of examination, extraction etc. The circumstances, to the extent they can be identified, have the potential to influence the way in which a court perceives the contamination. Close examination of the sequence and circumstances of collecting, transporting, testing, the nature of the items from which samples are collected and whether the logic of the situation would suggest if one sample was contaminated another or other samples would also have been contaminated may point towards or against the real, as opposed to theoretical, possibility of contamination.¹¹⁸ It does not follow that DNA evidence must be inadmissible if it is shown to have been contaminated. It will ultimately depend on the circumstances of the case. Where the court accepted the relevant DNA evidence extracted from a sock was contaminated by the DNA of a prisoner, who was not in prison at the time the prisoner / victim was murdered, the evidence was nevertheless admitted. On appeal held the relevant likelihood ratio was calculated on the basis the contaminated DNA was present and on the understanding it "contaminated" the DNA analysis. It did not lose its probative effect nor necessarily prevent the jury from properly evaluating the DNA evidence as a piece of circumstantial evidence. Redlich JA observed:¹¹⁹

The argument that the applicants had been deprived of a chance of demonstrating that they were excluded as contributors by virtue of the contamination must also be rejected. It involves the misconception that the likelihood ratio cannot be applied when there is a mixed DNA sample, in which the profile of other contributors could mask the presence of the suspect's profile, because there are common alleles between other contributors and the suspect at various sites.

Whether or not Ali's DNA material was present as a consequence of contamination, there was no suggestion that the analysis undertaken or the calculations were unreliable. The question whether there was contamination of the DNA material and if so, whether it had any significance, was left properly to the jury as a jury question.

In May 2010 Vincent J, recently retired from the Victorian Supreme Court, completed his inquiry into the circumstances that led to Farah Jama's wrongful conviction. Vincent J observed:¹²⁰

The DNA evidence appears to have been viewed as possessing an almost mystical infallibility that enabled its surroundings to be disregarded. The outcome was, in the circumstances, patently absurd. In Jama's case there is a very strong probability the contamination arose as a consequence of environmental factors at the point where a vaginal swab was taken by a doctor ...

In *Jama* there was a very strong probability the contamination arose when a vaginal swab was taken by a doctor from a woman in the Emergency Department of a hospital after she had been found unconscious in the toilet cubicle of a nightclub. Coincidentally approximately 28 hours earlier the same doctor, at the same location, had taken a forensic sample from another woman who had had sexual contact with Mr Jama.¹²¹ Testing a crime scene sample involves looking for "foreign material" in the sense that the test is seeking to find DNA that would not be present unless deposited as a consequence of the alleged offence. Often the crime scene sample involves a mixture of DNA from two or more sources (people) and, depending on the nature of the alleged crime, this may be inevitable. It is only by understanding the environment and processes surrounding the collection, transporting, testing etc of the crime scene and reference samples that a distinction can be made between a "true finding" and a "finding brought about by inadvertent contamination". Mr Jama's case was very unusual in that the only evidence against him was the DNA evidence comprising his DNA from one intact sperm cell and fifteen heads observed on only one of the four vaginal swabs taken from the woman found unconscious at the night club.¹²² All of the other prosecution evidence suggested it was unlikely a rape had occurred and, if it had, unlikely Mr Jama could have perpetrated it.¹²³

Guidelines to be applied when the admissibility of DNA testing is challenged have been suggested.¹²⁴ A correct ruling on the objection was said to depend upon identifying the precise basis of the challenge (ie danger of unfair prejudice over probative value, unfairness etc) and the precise testing step being challenged.

Where the objection is focused on the match results, rather than their statistical interpretation, then:

- given the courts treat evidence of scientific matching as an acceptable scientific technique, the probative value of properly qualified opinion evidence will be substantial; and
- the existence of a real risk of unfair prejudice must be assessed in the light of the:
 - acceptance a jury is capable of assessing such scientific evidence, even when presented with conflicting expert opinions; and
 - requirement the jury be given proper and careful direction on the evidence.

Where the objection is focused on the statistical interpretation of the match results then the:

- question of probative value and any risk of prejudice must be decided on the available evidence as to the validity of the statistical databases relied upon; and
- onus rests on the prosecution to provide sufficient and properly qualified evidence of the statistical validity of the databases relied upon.

As the Vincent report into *Jama's case* highlights it is the collection, transporting, storage and testing of both crime scene and reference samples that must be considered. When a match is made from the existing database an evidentiary reference sample will be required for the purposes of proof and this is done after the initial match between the crime scene sample and the database has been undertaken. Notionally if testing of the reference sample confirms the match between the crime scene sample and the database records it would be unlikely there has been contamination of the reference sample. However, in *Jama's case* it was the circumstances which gave rise to the initial reference sample being requested that pointed clearly to the contamination.

The protocols developed by the Victoria Police Forensic Services Department (VPFSD) and the Victorian Institute of Forensic Medicine (VIFM) regarding the collection, storage and testing of DNA samples are not publicly available. At the very least an appropriate level of transparency for the DNA component of a criminal prosecution requires a comprehensive index or catalogue of the protocols be available for public access on their respective websites. Prosecutors and defence lawyers need access to these protocols in appropriate cases and they are susceptible to subpoena.¹²⁵ Without a comprehensive index of protocols the subpoena process is unnecessarily cumbersome and inefficient use of court and legal resources. Conversely, it is clearly not appropriate to use up scarce forensic resources chasing rabbits in empty burrows.¹²⁶ If the issue in a sex case is consent it is highly unlikely DNA results confirming sexual contact can be an issue. If the accused admits being present at the crime scene or handling the offending weapon in the lead

up to the alleged offence it is unlikely the DNA evidence does any more than confirm an aspect of the defence and any competing hypotheses as to when and how DNA got there is a matter for the jury.¹²⁷ On the other hand if the defence is “I was not there” or there is potentially credible alibi evidence any DNA results incriminating the accused loom large and require careful testing. As always the available evidence in a case and an accused person’s instructions will determine whether it is necessary to challenge the DNA evidence and obtain access to the relevant protocols, notes etc.

See also discussion of contamination and admissibility at [\[ID.3280\]](#).

[ID.3260] Presenting the significance of the results

As noted above DNA evidence is given probative value by the statistical analysis.

The total exclusion of “arithmetical figures” skews the experts' evidence without contributing to the fairness of the trial. The prosecution case might be prejudiced: but no contribution would be made to the avoidance of unfair prejudice to the defence case.¹²⁸

There are different ways of expressing the results of comparing DNA from a crime scene sample with DNA from a reference sample, often from an accused.¹²⁹

Initially the “product rule” was used as a means of estimating how often a particular DNA profile will occur (sometimes referred to as the “random occurrence ratio”, being the frequency with which that DNA profile is likely to occur in the population). The product rule takes the frequency of individual DNA types at each locus in a profile where there was a match and multiplies it with the frequency of each other locus where there was a match – thus matches at three loci with a frequency of 1 in 7 for each locus will result in a profile frequency of $7 \times 7 \times 7 = 1$ in 343.

Criticism of the product rule, on the basis it assumed the population behaved like an ideal population rather than a population where people often choose partners from their own ethnic group, resulted in the adoption of “equation 4.10” (sometimes referred to as the “sub-population model”).¹³⁰ Equation 4.10 is a modification of the product rule to correct for the probability that two people chosen at random from the population share the same DNA profile. This gives a “match probability” – that is the probability a person chosen at random has that DNA profile given a known second person (say the accused) has that profile. It provides, for example, an estimate of the probability the DNA deposited by an unknown offender at the crime scene would match the DNA profile of the accused, given the accused says he could not be the source of the DNA. Using “match probability”, the smaller the probability the greater the likelihood the two samples came from the same person.¹³¹ Currently the sub-population model is used in Victoria.

The match probability is often expressed as a “likelihood ratio”, the ratio between two probabilities. The “likelihood ratio” is commonly used in Australia. Using “likelihood ratio”, the larger the probability the greater the likelihood the two samples came from the same person. Generally the two relevant probabilities are calculated by considering two alternative propositions to explain the evidence: the accused left the DNA at the crime scene (prosecution hypothesis) compared to someone other than the accused left the DNA at the crime scene (defence hypothesis). The likelihood ratio is calculated by dividing the match probability if the prosecution hypothesis were true by the match probability if the defence hypothesis were true.¹³² It is now accepted that the likelihood ratio is a valuable tool with the capacity to assist a jury.¹³³ Likelihood ratios are considered particularly useful in complex cases such as DNA mixtures and paternity cases.¹³⁴

In 2013 the VPFSS (Victorian Police Forensic Science Service) introduced the STRmix program as a new tool for calculating the likelihood ratio. The STRmix, a statistical software program developed by Australian and New Zealand scientists, has been adopted by New Zealand and all Australian States and Territories except Tasmania and the ACT. The STRmix program is seen as having particular benefit in circumstances where the sample involves low-template or low copy number (LCN) DNA profiles. Analysis of such samples will often involve stochastic (random) effects, see discussion at [ID.3160] “Trace DNA – Low copy number typing”. The STRmix program is intended to take account of the random effects in a DNA profile because it does not apply an allele peak threshold below which peaks will not be interpreted. It involves the continuous modelling of peaks heights and, among other things, purports to take into account peaks that are not detected by positing that the alleles have “dropped-out” of the profile and giving them a value. Applying a further tool, the Markov Chain Monte Carlo (MCMC), the STRmix identifies genotype combinations that best explain the observed DNA profile. The STRmix produces a comprehensive list of possible explanations for a profile and ranks those explanations according to how well they explain the profile obtained. The reference sample, say of the accused, is compared to the weighted list of genotype combinations. If the reference sample genotypes are among the highest weighted genotype combinations in the sample (crime scene) to which the STRmix software has been applied then a significant likelihood ratio will be produced. This approach means a person, the accused, will not be excluded as a possible contributor to the crime scene sample even though the accused’s alleles were not detected at a particular allele in the crime scene sample if the STRmix has posited a ‘drop-out’ at that marker. Rather the absence of the accused’s alleles at that site will result in a decreased likelihood ratio, but provided the likelihood ratio is ‘significant’ the accused is not excluded as a possible contributor to the crime scene sample. If it is not a significant likelihood ratio the person in the reference sample (the accused) will be excluded as a contributor to the crime scene sample. A more complete description of the methodology applied by STRmix is contained in *R v Tuite* [2014] VSC 662.¹³⁵ Whereas earlier methods of calculating likelihood ratios used a binary system, essentially a match or a non-match, the STRmix creates a number of possible genotype combinations that best explain the observed DNA profile and weights them using the

MCMC tool. Significantly, this means STRmix produces an outcome that will not be reproduced if the process is repeated as the random nature of the MCMC tool will produce slightly different outcomes each time. Consequently the degree of expected variation in the results is compared to the actual variation achieved by the results. If the actual variation is within the expected variation the information obtained is used.

Whether STRmix satisfied the requirements of [s 79\(1\) of the Evidence Act 2008](#) was considered in *R v Tuite* [2014] VSC 662. Emerton J concluded:¹³⁶

Although neither article endorses STRmix specifically, or the fully-continuous methodology that it uses more generally, the articles place the fully-continuous probabilistic statistical methodology within the evolving field of knowledge concerned with the statistical evaluation of DNA evidence. This is not to say, however, that there may not be problems associated with fully-continuous systems or that particular DNA scientists might well prefer other models or methodologies.

... this evidence places STRmix and its fully-continuous probabilistic methodology squarely within the field of statistical DNA evaluation, albeit possibly at its leading edge. It has a credible scientific and mathematical basis. However, that is not to say that the results that it produces are inherently reliable. It may not produce results that are as reliable for the purposes of forensic case-work as binary or semi-continuous methods. Its method for determining the possibility or probability of drop-out based on peak height variability may be open to criticism. Its use on compromised profiles may be questionable. But those are matters for competing expert opinion, providing, of course, that STRmix is amenable to scrutiny and independent testing.

...

On the basis of the evidence at the preliminary hearing, I have concluded that knowledge concerning the statistical evaluation of DNA profiles, including by a fully continuous probabilistic system such as STRmix, is “specialised knowledge” that could support opinion evidence in the form of likelihood ratios generated by STRmix. The DNA evidence proposed to be given by Ms Scott is admissible as opinion evidence under [s 79\(1\) of the Evidence Act](#) having regard to the supporting evidence of Ms Federle and Dr Taylor

An alternative method, not currently used in Victoria, is the exclusion probability or the Random Man Not Excluded (RMNE).¹³⁷

Many difficulties have been experienced in relation to the proper means of statistically interpreting the results of DNA profiling tests. A key issue is the size and racial spread of the databases against

which the sample of the accused is compared. In *R v Pantoja*¹³⁸ Hunt CJ at CL and Hidden J held that until a general acceptance is accorded to the appropriate size of DNA databases, the Crown should lead such evidence where objection is taken to the admissibility of the statistical interpretation of the DNA match. Such databases must be shown to be statistically valid. If the identity of the offender may have been a person from a small population sub-group, a direction should be given to the jury that:¹³⁹

[T]he chance of a coincidence of a match thrown up by the use of such databases would have to be reduced in order to take that possibility into account.

Abadee J held that if the accused belongs to a particular racial group or sub-group the preparation of a specific database rather than a general database:¹⁴⁰

[M]ay well be warranted. ... [I]t may be that there will be a need for preparation of several different likelihood ratios or databases.

However, in *R v Karger*¹⁴¹ calculation via “the product rule” in which results from different loci are multiplied one with another and via *Equation 4.10* could be undertaken using databases of modest dimensions, the inferences to be drawn from the results being the province of the jury.¹⁴²

The need for confidence limits to be employed in the statistical interpretation of DNA profiling results remains somewhat unclear. However, Hunt CJ at CL has held that the jury in every DNA case should be directed not to approach the issue of chance “upon any strictly mathematical basis”.¹⁴³ Of itself, this is insufficient. It is necessary for the jury to be provided guidance on how to approach the issue. The jury must assess the significance of DNA evidence in the context of all of the evidence and not allow it to assume undue importance in their deliberations regardless of how great the statistical likelihood asserted by the prosecution.

In the United Kingdom in relation to a no case submission it has been held:¹⁴⁴

Every case of this kind has to be judged on its own facts. There is no rule that enables the court to say, well, when a figure reaches a certain level then it is safe to leave it to the jury, but below that it is not. But in every case one has to put the DNA evidence in the context of the rest of the evidence and decide whether taken as a whole it does amount to a *prima facie* case.

This approach is equally applicable, if not more so, where the likelihood ratio relied upon by the prosecution involves numbers in the millions and billions – its significance must be assessed in light of all the available evidence.

On the ultimate question of guilt, evidence of this magnitude is evidence that the accused is one of a very small number of people in the population from whom the DNA profile could have come. The issue for the jury is whether it can be satisfied it was the accused who left the crime scene sample from which the DNA was obtained. It may be the jury is able to conclude, given the very small number of people from whom the profile could have come, the DNA evidence is strong circumstantial evidence the accused left the crime scene sample. Even if the jury is so satisfied it must still be satisfied on all of the evidence beyond reasonable doubt that the accused is the person who committed the acts alleged and if it reaches that level of satisfaction that the alleged offence is made out beyond reasonable doubt. Depending on the facts of the case this may be one and the same thing.¹⁴⁵

The closer an expert witness's evidence comes to expressing an opinion that bears upon the ultimate issue for the tribunal of fact the greater the need for scrutiny by the court. The court must be careful the witness does not move beyond their area of expertise or express views not supported by the disclosed and contestable assumptions.¹⁴⁶ Care must be taken when evidence, argument or directions are presented to a jury or tribunal of fact to avoid falling into the "prosecutor's fallacy". The prosecutor's fallacy runs thus – only one person in a million has a DNA profile that matches the crime scene sample's DNA profile, the defendant's DNA profile matches the crime scene profile and therefore there is a million to one probability the defendant left the crime scene sample and is guilty of the crime. The error of this reasoning was demonstrated in *Doheny v The Queen*:¹⁴⁷

Taking our example, the prosecutor's fallacy can be simply demonstrated. If one person in a million has a DNA profile which matches that obtained from the crime stain, then the suspect will be 1 of perhaps 26 men in the United Kingdom who share that characteristic. If no fact is known about the Defendant, other than that he was in the United Kingdom at the time of the crime the DNA evidence tells us no more than that there is a statistical probability that he was the criminal of 1 in 26.

In *Ali v The Queen* [2022] VSCA 31 the Victorian Court of Appeal observed:¹⁴⁸

The prosecutor's fallacy involves the 'transposition of the conditional'; that is, it takes a statement about the probability of one thing (X) relative to another (Y), and inverts it to a statement about the probability of Y relative to X. In the DNA context, the prosecutor's fallacy takes the 'random match probability' — in the present case, the probability of a DNA match if the defendant was the (or a) source of the DNA (which is evidence that an expert is

properly able to give) — and inverts it to a probability that the defendant is the (or a) source, given the DNA match (which is not evidence an expert is properly able to give).

Reference has also been made to the “defence attorney’s fallacy”, namely that associative statistical evidence is irrelevant regardless of the rarity of the matching characteristics. Of this assertion it has been said:¹⁴⁹

What this reasoning fails to take into account is that the great majority of the pool of people with relevant matching are not suspects in the case against whom additional evidence is available linking them with the offence.

The use of an “exclusion percentage” to describe the statistical result has been argued to be unfairly prejudicial to the accused. An exclusion percentage is the percentage of the population who are excluded as possible contributors to the relevant crime scene sample based on the frequency with which the DNA profile in question would be expected to be found within the particular population database used to calculate the frequency ratio. Put another way, the exclusion ratio expresses the frequency ratio in the reverse. The exclusion ratio was also said to be irrelevant on the basis it could not, in any significant way, rationally add anything to the jury’s assessment of guilt given evidence of the frequency ratio was before the jury.¹⁵⁰ The exclusion ratio in question was 99.9%, based on a frequency ratio of 1 in 1600. The plurality of the High Court rejected both arguments. The evidence was not unfairly prejudicial, misleading or confusing given the expert witness explained both the frequency ratio and the exclusion percentage and how they related to each other. The judge made clear to the jury the evidence of mtDNA profile found in the hair did not establish the hair came from the accused.¹⁵¹ Further, the accused accepted the evidence of the exclusion percentage was relevant in that it could rationally affect the probability of the existence of a fact in issue and its probative value was therefore necessarily the same as that of the frequency ratio given they were different ways of expressing the one statistical statement.¹⁵² The plurality allowed there may be cases:¹⁵³

... where evidence given of exclusion percentages may warrant close consideration of the application of ss 135 or 137 [of the *Evidence Act 1995 (NSW)*] (or, where applicable, equivalent common law doctrines or statutory provisions). These reasons are not to be read as suggesting to the contrary. Evidence given about the results of DNA analysis is evidence about comparisons between identified samples and one or more databases. The results of those comparisons can be expressed qualitatively or quantitatively. If expressed quantitatively there are assumptions and approximations made which often (perhaps always) require elucidation and explanation to make plain what are the limits to the opinion that is being expressed as a number or range of numbers. Just as evidence of an opinion given by an expert must, in order to satisfy the requirements of admissibility in s 79(1) of the

Evidence Act 1995 (NSW), be “presented in a form which makes it possible to answer” the question posed by that provision, it will usually be important, even necessary, that the evidence provides the jury with so much of the expert’s “specialised knowledge” as the jury requires properly to understand the opinion expressed — and what it can and cannot demonstrate — and that this specialised knowledge be related to the facts of the case.

It may very well be right to observe that a frequency ratio of one in 1,000 can, even may often, convey a different message to the hearer than does an exclusion percentage of 99.9%, because the denominator of the frequency ratio directs the hearer’s attention to the population that must be considered when seeking to apply the ratio. (Of course, a percentage, too, is a ratio but may direct less attention to the denominator.) Not only that, it is important to recognise that evidence of DNA analysis tendered by the prosecution is tendered in proof of a case that the accused is guilty of the offence charged. It is not usually tendered only to exclude the possibility that there may be others who committed the offence (unless the possible class of offenders is limited). It is usually tendered to show that there is at most a small pool of persons, including the accused, who could have left a trace at the scene of the crime. But demonstrating that there are many persons in Australia who did not commit the crime charged against the accused may be thought, if that information is considered in isolation, to tend to distract attention from whether the accused is the one out of the remaining number of possible perpetrators who did commit the crime.

The manner in which the significance of the results of DNA analysis is presented to a jury is a matter that requires careful consideration by the parties and the court. Expert witnesses use different methods, including slides and graphs to assist a jury to understand the technical explanations they are providing. These difficulties were considered in *Vyater v The Queen* [2020] VSCA 32 where, after setting out extracts from the expert witness’ evidence to the jury of the DNA Analysis procedure, the Court of Appeal observed: ¹⁵⁴

Having reviewed the slides together with the oral evidence, however, we found the technical explanations difficult to understand and we would assume that jurors would have similar difficulties. This prompts us to question whether it is either realistic or necessary to expect a jury to understand the science of DNA profiling, as distinct from the ‘evaluative opinion’ conveyed as a likelihood ratio or its verbal equivalent.

In *Vyater* [2020] VSCA 32 the Court also considered the expert witness’ evidence, given during cross-examination, explaining why a particular likelihood from a glove was so low in circumstances where the likelihood ratio from a safety mask was 100 billion to one. In regard to that part of the expert witness’ evidence the Court of Appeal observed: ¹⁵⁵

In our respectful opinion, this explanation — while scientifically orthodox — would also have been very difficult for a lay juror to understand. It required an appreciation of the process of cross-multiplication' of the various likelihoods of occurrence at the various sites in the DNA profile and, more particularly, of how the result of that 'multifactorial' process can be affected both by the number of available sites and by the relative (lack of) frequency of particular DNA types. Even if this technical explanation were capable of being understood, we doubt that it would have assisted a juror to make sense of the difference in probative force between a likelihood ratio of 26 to 1 and a ratio of 100 billion to 1.

Despite these concerns regarding the perceived difficulties the jury would have had understanding the expert evidence the Court of Appeal concluded there was little risk of the disputed low likelihood ratio evidence being overvalued by the jury given the exemplary directions given by the trial judge. The Court concluded that given the other circumstantial evidence relied on by the prosecution the probative value of the contentious DNA evidence was high. This was so in the context of the evidence because the description it was 'distinctly more probable' the accused contributed to the DNA profile taken from the glove in issue was justified and provided 'moderate support' for the prosecution hypothesis even though the likelihood ratio for that DNA match was low. The reasons of the Court reinforce the significance and probative value of DNA will depend heavily on other circumstantial evidence, other evidence about the accused and the approach taken by an accused to the prosecution case. ¹⁵⁶

Where the paternity of a child is a relevant matter to the resolution of the alleged crime the results of DNA matching can be expressed as a "paternity index", being the odds in favour of paternity. It is expressed in terms such as "it is 30,000 to 1 times more likely D is the biological father of C than any other male taken at random in the population". Alternative ways of expressing the paternity index, such as a percentage or the "relative chance of paternity" have been rejected as more difficult to explain to a jury and likely to be unduly prejudicial to an accused in its impact. ¹⁵⁷

[ID.3280] Can an accused be convicted on DNA evidence alone?

Whilst there is absolutely no bar to a conviction based solely on DNA evidence, the better view is that a conviction should only be returned where there is DNA evidence and at least one other item of evidence present which is consistent with the guilt of the offender. ¹⁵⁸

Evidence of a match between the DNA of the offender (the crime scene sample) and the accused establishes no more than the accused:

- *cannot be excluded* as the person responsible for the crime scene sample; or put another way
- *could be* the offender.

It cannot, in the absence of other evidence, prove beyond reasonable doubt the accused was the person responsible for the crime scene sample.¹⁵⁹

The difficulty defence lawyers face where the prosecution evidence consists entirely of a DNA match, is that any evidence adduced by the defendant before the jury, including alibi evidence, means the case is no longer a case where the only evidence for the jury to consider is the DNA evidence. The Vincent report into *Jama's case* noted Mr Jama did not give evidence, relied upon the record of interview tendered by the prosecution and called several alibi witnesses, who it was said fared badly under cross-examination.¹⁶⁰ In *Forbes v The Queen*,¹⁶¹ where DNA evidence was the only evidence supporting the conclusion the accused was the perpetrator, the accused gave evidence and called his wife in support of his alibi that he was at home at the time of the alleged rape. Notably in *Jama's case* the defence conceded there was no issues of contamination of the DNA and contended the DNA must be from an unidentified third person.¹⁶² In *Forbes v The Queen*,¹⁶³ whilst the DNA evidence was tested at trial, it was not suggested the DNA was contaminated or that there were problems with the testing. The focus in *Forbes v The Queen* was on the limitations inherent in probability ratios given only 10 locations were tested and the results are only a likelihood ratio, not evidence it was the accused's DNA found at the crime scene.¹⁶⁴ The problem is that whilst a disbelieved alibi or the rejection of other defence evidence cannot be proof of the alleged offences, they are matters a jury is entitled to take into account when assessing all of the evidence to determine whether they are satisfied beyond reasonable doubt the prosecution has made out the requisite matters of proof.¹⁶⁵

Fundamental principles provide appropriate guidance. Evidence of a DNA match between an accused person's DNA and a crime scene or related sample is circumstantial evidence. If there is no dispute as to the DNA match, the question must be how the accused's DNA got to be on the crime scene or related sample. The prosecution must exclude any reasonable hypothesis consistent with the accused's innocence. In a case where the only evidence against an accused is DNA evidence, this is the only "link in the chain" of circumstantial evidence. If the prosecution cannot exclude the possibility of secondary transfer of the accused's DNA to the crime scene or related object it is difficult to see how the prosecution could possibly exclude any reasonable hypothesis consistent with innocence. Locard's principle, with every contact between two items there will be an exchange, makes it difficult to exclude the possibility of secondary transfer.¹⁶⁶ It is often other evidence implicating an accused person that allows the possibility of secondary transfer or contamination of DNA to be excluded as a reasonable hypothesis. Where the only evidence is DNA evidence a rigorous examination of the circumstances surrounding the crime scene or related samples location, collection etc is critical to the issues of secondary transfer and contamination. Whilst not strictly a case where the only evidence was DNA, a clear example of the degree of rigor required by both prosecution and defence when considering the possibility of secondary transfer can be found in the reasoning of the High Court *Fitzgerald v The Queen*.¹⁶⁷ In *Fitzgerald [2014]*

HCA 28 the prosecution relied upon DNA of the accused obtained from a didgeridoo found in close proximity to where the deceased was left after being attacked. The accused's DNA was contained in one of five samples taken from the didgeridoo. There was unchallenged evidence, given by the co-accused Sumner, of physical contact in the form of a handshake between the accused and a co-accused earlier on the day of the attack. The prosecution forensic expert gave evidence there were 3 possibilities for the transfer of blood to an object, direct, airborne and passive transfer. She also gave evidence of the difference between primary and secondary transfer where secondary transfer involved transfer of DNA onto an object by way of an intermediary, such as with a handshake. In circumstances where the only evidence implicating the accused is a DNA match, if the accused can demonstrate there is a real possibility of contamination then, unless the prosecution can demonstrate it is not contaminated, the evidence may be excluded on the basis its probative value is exceeded by its prejudicial effect. A real possibility is one that goes beyond a theoretical possibility and substantially erodes the probative value of the evidence.¹⁶⁸ Ultimately, if the accused points to a real possibility of contamination, the onus rests on the prosecution to establish beyond reasonable doubt the DNA evidence is not contaminated or that even though the possibility of contamination exists it is a proper matter for the jury to determine. Given it is difficult to establish a negative, the obligation on the prosecution is more aptly described as establishing there is no evidence of contamination. If the prosecution does not satisfy one of these requirements the evidence is inadmissible. Whilst this sometimes may apply even where DNA evidence is just another part of the prosecution case, it assumes fundamental importance where the only evidence capable of implicating the accused is DNA evidence. In cases where DNA is only just another piece of the prosecution evidence the defence often make concessions regarding DNA evidence, but concessions can be appropriately made only if the defence have carefully considered all aspects of the DNA evidence as a discrete issue. In his report into *Jama's case* Vincent J comments:¹⁶⁹

So far, Australian courts have been extremely reluctant to engage in the testing of the admissibility of DNA evidence and its possible exclusion by reason of its unduly prejudicial effect through the use of voir dire procedures under which the judge determines these questions, leaving it to the jury to struggle with the problem and with little assistance in dealing with it. That is, in my view, most unsatisfactory and should be addressed.

If there is no evidence to suggest the possibility of contamination and no other evidence for a jury to consider then the logic of the assertion a DNA match in the absence of other evidence cannot prove beyond reasonable doubt the accused was the person responsible for the crime scene sample, invites a no case submission.¹⁷⁰

Apart from demonstrating the rigor required when assessing DNA evidence the reasoning of the High Court in *Fitzgerald's case* [2014] HCA 28¹⁷¹ suggests, without addressing the issue directly, the answer to the question whether there should be a rule of law a conviction cannot

based solely on DNA evidence is probably “no”. Rather the requirements for a conviction based on circumstantial evidence must be rigorously applied and the prosecution must satisfy the court any reasonable possibility of secondary transfer or contamination have been excluded in addition to any other reasonable hypothesis consistent with innocence. A prosecuting authority should not proceed with a case based on DNA evidence alone unless it is satisfied it can meet that test. The defence must be prepared to make every available challenge to the DNA based on the possibility of secondary transfer, contamination and any other possible hypothesis consistent with innocence. Given as noted above DNA cannot, in the absence of other evidence, prove beyond reasonable doubt the accused was the person responsible for the crime scene sample it is difficult to see how all other reasonable hypothesis consistent with innocence can be excluded if the requirements for a conviction are applied rigorously.

The nature of the DNA in the particular case is clearly important. In both *Jama's case* and *Forbes v The Queen* the results were from autosomal DNA. If the only evidence implicating an accused is mtDNA it is difficult to see how a prosecution could proceed given the lack of discrimination implicit in mtDNA. In those circumstances the mtDNA only establishes the accused has the same mtDNA as that found at the crime scene – it does not even establish whether the person who deposited the crime scene sample was male or female. The same lack of discrimination applies to Y chromosome DNA except Y chromosome DNA can only come from a male.

The English courts hold there is no rule of law that DNA evidence alone is incapable of proving guilt. In *R v Adams (No 1)* ¹⁷² Rose LJ observed:

None of these provides any support for Mr Thwaites' proposition that DNA evidence alone is incapable of proving guilt. That is unsurprising. In every case, the probative effect of a particular piece of evidence depends on its quality individually and in relation to all the other evidence. It is, in our view, wrong to describe DNA evidence generally, or the DNA evidence that was presented in this case, as no more than a rough estimate of that which it seeks to establish.

Any of the figures mentioned, taken on their own, are plainly very cogent evidence. However, whether the evidence of those figures is reliable in itself depends on scrutiny of the assumptions and methodology of the original calculations, which can be the subject of enquiry at trial. The extent to which such evidence survives such scrutiny sufficiently to establish guilt is, as we have said, a separate question to be assessed in the light of all the evidence in the case. There is, however, nothing inherent in the nature of DNA evidence which makes it inadmissible in itself or which justifies a special, unique rule, that evidence falling into such a category cannot found a conviction in the absence of other evidence.

It can be accepted this will ultimately prove to be the position in Australia.

[ID.3300] Directions to the jury

DNA identification evidence is not identification evidence as defined by the *Jury Directions Act 2015*.¹⁷³ Expert evidence is not given any specific treatment in the *Jury Directions Act 2015* and is not a matter that comes within the definition of “general directions”.¹⁷⁴ It follows directions on expert evidence are subject to the requirements of Part 3, “Request for directions”, of the *Jury Directions Act 2015*. Even where there is no dispute about DNA analysis evidence, it may be an agreed fact between the parties for example, the evidence may still be a matter in issue in the trial. For example, what inferences are to be drawn from the agreed DNA evidence may be in issue and be the subject of specific direction from the trial judge. The required directions, if any, will be informed by the role DNA evidence has played in the trial and the way the parties have treated that evidence. DNA evidence does not have to be proved beyond reasonable doubt necessarily. Pursuant to the *Jury Directions Act 2015* the trial judge may only direct the jury they must find proved beyond reasonable doubt the elements of the offence or any alternative offence.¹⁷⁵

The trial judge must exercise considerable care in formulating the way in which the DNA evidence is summarised to the jury. It should be put to the jury that the evidence, if accepted, means that the accused cannot be excluded as the perpetrator and that, therefore, it is possible that he or she was the person responsible for the crime scene sample.¹⁷⁶ A match obtained by any blood tests, DNA or otherwise, between the suspect and the offender, does not establish that the two are one and the same person; it establishes no more than that the accused could be the offender.¹⁷⁷ The situation is very different should the DNA evidence positively exclude the accused as the offender. In those circumstances it may be necessary to instruct the jury that, if they accept there is any reasonable possibility the test is correct, that must exclude the accused notwithstanding other evidence.¹⁷⁸

Where DNA evidence is an indispensable link in the chain of reasoning to establish a fact necessary to support a conviction the reliability of the DNA evidence must be established beyond reasonable doubt. If the DNA evidence is part of the circumstantial evidence in a case resting on the accumulation of detail, or the strands in a cable, it is not appropriate to instruct a jury they must be satisfied of the reliability of the DNA evidence beyond reasonable doubt before they can draw the ultimate inference of guilt.¹⁷⁹ In the latter situation, consistent with the general principles relating to circumstantial evidence, it is open to the jury to be satisfied beyond reasonable doubt on the totality of the evidence.

Expert evidence does not require a specific direction of law to the jury. Rather:¹⁸⁰

[T]he overriding duty of the trial judge to give such directions as are necessary to avoid any perceptible risk of a miscarriage of justice remains. In the case of DNA evidence, the content of the charge must adequately address the danger that the jury not attribute to the

DNA evidence a weight that is unjustified because of the complexity of the evidence and the form in which the scientific opinion is expressed. This will ordinarily be achieved by a careful summary of the expert evidence which includes the limited nature of the conclusions reached.

[ID.3320] Guidelines where DNA evidence is to be adduced

The Northern Territory Court of Criminal Appeal in 1998 provided the following guidelines where the Crown proposes to lead DNA profiling evidence: ¹⁸¹

- (1) Whenever DNA evidence and statistical evidence based thereon is to be adduced, the Crown should serve on the defence prior to the committal hearing a statement or statements from the expert or experts the Crown intends to call, which provides details of the DNA testing carried out, the nature of the matching DNA characteristics between the DNA in the crime sample and the DNA obtained from the defendant, and details as to how the calculations of the likelihood ratio have been carried out which are sufficient for the defence to scrutinise the basis of the calculations.
- (2) Provided that the expert has the necessary data, it may then be appropriate for it to be indicated how many people with the matching characteristics are likely to be found in Australia, or in a more limited sub-group, for instance, the sexually active males in the Darwin area, depending on the circumstances of the case.
- (3) If the Crown intends to supplement or change the DNA evidence or the statistical evidence based thereon, after the committal hearing, it should serve such additional statements as are necessary to comply with guideline (1) in sufficient time prior to the trial for the defence to be able to meet that evidence.
- (4) The forensic section of the Northern Territory Police Department should make available to a defence expert, if requested, the databases on which the calculations have been based (but not information which identifies particular individuals included in the databases). Any failure to do so in time for the defence expert to be available to assist the defence at the trial may lead to the exclusion of any statistical evidence at the trial.
- (5) Wherever possible, sufficient of the crime scene sample should be kept by the forensic section of the Northern Territory Police Department for re-testing, and made available to the defence for that purpose, upon request.
- (6) It is not necessary for the Crown to lead evidence from an expert in population genetics or from another scientific expert as to the statistical validity of the databases kept by the forensic section of the Northern Territory Police Department where the defence notifies the Crown that this is not in issue, or where objection is not taken at trial.

- (7) A scientist other than a population geneticist or an expert in a statistical discipline may have sufficient qualifications derived from professional experience and personal familiarity with the data on the relevant database and published population statistics to be permitted to give evidence of the likelihood ratios in the relevant population. If the Crown proposes to adduce evidence of this kind from such a scientist, the Crown should serve on the defence in accordance with guidelines (1) or (2) a statement of the scientist's qualifications and experience.
- (8) Disputes as to the admissibility of DNA and statistical evidence, including the qualifications of witnesses, should be determined wherever possible by utilising the procedure provided for in s 26L of the *Evidence Act 1939* (NT).
- (9) Experts called to give statistical evidence should be led by the Crown as to any assumptions made in their calculations which, even though widely accepted, are not supported by empirical research, including:
 - (a) Hardy-Weinberg equilibrium;
 - (b) where the offender is of a racial group or sub-group for which there is no valid database and a general database has been used which does not take that fact into account, that fact.
- (10) Experts should not give evidence as to the likelihood that it was the defendant's DNA found at the crime scene or use terminology suggesting that he or she is expressing such an opinion.

The guidelines were in part derived from those proposed by the English Court of Appeal in *Doheny v The Queen*:¹⁸²

1. The scientist should adduce the evidence of the DNA comparisons between the crime stain and the defendant's sample together with his or her calculations of the random occurrence ratio.
2. Whenever DNA evidence is to be adduced, the prosecution should serve on the defence details as to how the calculations have been carried out. The calculations should be sufficient to enable the defence to scrutinise the basis of the calculations.
3. The Forensic Science Service should make available to a defence expert, if requested, the databases upon which the calculations have been based.
4. Any issue of expert evidence should be identified and, if possible, resolved before trial. This area should be explored by the court in the pre-trial review.
5. In giving evidence, the expert should explain to the jury the nature of the matching DNA characteristics between the DNA in the crime stain and the DNA in the defendant's blood sample.
6. The expert should, on the basis of empirical statistical data, give the jury the random occurrence ratio – the frequency with which the matching DNA characteristics are likely to be found in the population at large.

7. Provided that the expert has the necessary data, it may then be appropriate for him or her to indicate how many people with the matching characteristics are likely to be found in the United Kingdom or a more limited sub-group, such as the caucasian, sexually active males in the Manchester area.
8. It is then, for the jury to decide, having regard to all the relevant evidence, whether they are sure that it was the defendant who left the crime stain, or whether it is possible that it was left by someone else with the same matching DNA characteristics.
9. The expert should not be asked his or her opinion on the likelihood that it was the defendant who left the crime stain, nor when giving evidence, should the expert use terminology which may lead the jury to believe that he or she is expressing such an opinion.
10. It is appropriate for an expert to expound a statistical approach to evaluate the likelihood that the defendant left the crime stain, since unnecessary theory and complexity deflect the jury from their proper task.
11. In the summing-up by the trial judge, careful directions are required in respect of any issues of expert evidence. Guidance should be given to avoid confusion caused by areas of expert evidence where no real issue exists.
12. The trial judge should explain to the jury the relevance of the random occurrence ratio in arriving at their verdict and should draw attention to the extraneous evidence which provides the context which gives that ratio its significance, and to that which conflicts with the conclusion that the defendant was responsible for the crime stain.
13. In relation to the random occurrence ratio, a direction along the following lines may be appropriate, tailored to the facts of the particular case: "Members of the jury, if you accept the scientific evidence called by the Crown this indicates that there are probably only four or five white males in the United Kingdom from whom the stain could have come. The defendant is one of them. If that is the position, the decision you have to reach, on all the evidence, is whether you are sure that it was the defendant who left that stain or whether it is possible that it was one of that other small group of men who share the same DNA characteristics".

As noted above since 1998/1999 all State and Territory forensic science laboratories have used the Profiler Plus as a standard testing procedure for DNA samples and although its admissibility has been challenged in a number of cases it has so far withstood those challenges. Supreme Court decisions in South Australia and New South Wales have held expert evidence deriving from testing using the Profiler Plus system to be admissible.¹⁸³

Body tissue sampling and physical examinations

[ID.3340] Definitions

The *Crimes Act 1958* differentiates between “intimate” and “non-intimate” parts of the body and between “intimate” and “non-intimate” samples. Other important distinctions include those between “forensic procedures” and “compulsory procedures” and between “procedures” and “physical examinations”.

An “intimate sample” is defined as: ¹⁸⁴

- (a) a blood sample;
- (b) a sample of pubic hair, including the root if required;
- (c) a swab, washing or sample taken from the external genitalia or anal region of a male or female or from the breast of a female;
- (d) a sample of saliva;
- (e) a scraping taken from the mouth;
- (f) a dental impression.

An “intimate part of the body” is defined as the “genital or anal region of a male or female or the breast of a female”. ¹⁸⁵

A “non-intimate sample” is defined as: ¹⁸⁶

- (a) a sample of hair, other than pubic hair, including the root if required;
- (b) a sample of matter taken from under a fingernail or toenail;
- (c) a swab, washing or sample taken from any external part of the body other than the genital or anal region of the male or female or the breast of a female.

A “non-intimate part of the body” is defined to mean any part of the body other than an “intimate part”. ¹⁸⁷

A dental impression is not defined.

A “physical examination” is defined as: ¹⁸⁸

... an examination of the external part of a person's body requiring touching of the person or removal of the person's clothing.

A “forensic procedure” is defined as:

... the taking of a sample from any part of the body, whether an intimate or non-intimate sample or any other type of sample, or the conduct of any procedure on or physical examination of the body.

However, it is expressed not to include a fingerprint, which in turn is defined to include a finger, palm, toe and sole print.¹⁸⁹

A “compulsory procedure” is defined to mean:¹⁹⁰

... the taking of an intimate or non-intimate sample or the conduct of a physical examination.

[ID.3360] Forensic procedures

A member of the police may request any adult to undergo:

... the taking of a sample from any part of the body, whether an intimate or non-intimate sample or any other type of sample, or the conduct of any procedure on or physical examination of the body.

However, even this power of request that a person undergo a “forensic procedure” is circumscribed. The request may only be made if there are reasonable grounds to believe that the procedure would tend to confirm or disprove the involvement of the suspect in the commission of an indictable offence. In addition, the suspected involvement of the person must be direct – by way of their being reasonably suspected of having committed the offence, or their having been charged with it or summonsed to answer a charge for the offence.¹⁹¹

A “forensic procedure” may be undertaken upon a suspect if the suspect gives his or her informed consent, the Magistrates' Court orders it to take place or a senior police officer gives an authorisation, in relation to a “non-intimate compulsory procedure”.¹⁹² The concept of informed consent is defined in terms similar to those in [s 53 of the *Mental Health Act 1986*](#) which deals with provision of consent to electroconvulsive therapy and psychosurgery. It requires substantial provision of information to the suspect in terms of why the investigative police wish the procedure to be undertaken, what the procedure is, how it is to be undertaken, how it can be used and what the suspects rights are.¹⁹³ The information must be provided to the person in language likely to be understood by the person. This is an objective test and does not require the police officer to establish the person actually understood the information given.¹⁹⁴ The failure of the police officer to provide some of the required information is not necessarily fatal to the admission of any evidence obtained as a result of the deficiency in the informed consent requirement.¹⁹⁵ In addition, police

are obliged to record their provision of the information and the responses, if any, of the suspect.¹⁹⁶ If the consent is obtained by a misrepresentation or by trickery, the admissibility of the evidence resulting from it will be discretionary on the same grounds as evidence of confessions elicited in the same way. Informed consent does not mean informed by legal advice, which forms no part of the requirements of s 464S.¹⁹⁷

[ID.3380] Court-ordered compulsory procedure

A “compulsory procedure” is defined by s 464 of the *Crimes Act 1958* to mean the taking of an intimate or non-intimate sample or the conduct of a physical examination.

A police officer can apply to a Magistrates' Court for an order directing a person to submit to a “compulsory procedure” if:

- the suspect has refused to submit; or
- the suspect is incapable of providing informed consent by reason of mental impairment; and
- the sample or examination sought may be obtained by a “compulsory procedure”; and
- the person is a “relevant suspect”;¹⁹⁸ and
- the police member believes on reasonable grounds that the person has committed the offence in respect of which the procedure was requested.¹⁹⁹

The court can make an order for submission to a compulsory procedure if satisfied of a number of matters on the balance of probabilities. First, it must be satisfied that the person is a “relevant suspect”,²⁰⁰ and secondly that there are reasonable grounds to believe that the person has committed the offence in respect of which the application is made.²⁰¹ These requirements recognise:²⁰²

[T]hat mandatory procedures have a potential to represent the antithesis of historic rights of citizens against self incrimination and the statute legislates requirements and limitations in order to strike a necessary balance between the appropriate use of available scientific means for investigating suspected crime and those rights.

It is the magistrate who must be satisfied on the balance of probabilities that the subject of the application is a “relevant suspect” and that there are reasonable grounds to believe that person committed the offence(s) in relation to which the application is made.²⁰³ It follows that there must be a specific offence(s) particularised in the application. It might be argued the magistrate must be satisfied to the required standard on evidence before the court and if the evidence is deficient the order cannot be made.²⁰⁴ It follows that the magistrate must be satisfied the subject of the application is a suspect not that the granting of the application and analysis of subsequent sample will allow the police to identify the subject person as a suspect.²⁰⁵

Save in an application to take a gunshot residue sample, it must also be satisfied that:

- material reasonably believed to be from the body of a person who committed the offence has been found at the scene of the offence, on the victim, on anything reasonably believed to be associated with the victim or on an object or person reasonably believed to have been associated with the commission of the offence; or
- that there are reasonable grounds to believe that because of the nature of the offence or injuries inflicted during the offence, material from the body or clothing of the victim is present on the offender or anything reasonably believed to have been worn or carried by the offender when the offence was committed or on an object reasonably believed to have been associated with the commission of the offence; or
- the victim of the offence had not been found and there are reasonable grounds to believe that material reasonably believed to be from the body of the victim is present on a person suspected of having committed the offence; or
- the offence is a sex offence and that there are reasonable grounds to believe that the procedure may demonstrate the paternity of a child allegedly conceived as a result of the offence.²⁰⁶

The court must also be satisfied that:

- in the case of a gunshot residue application for a sample or washing from the skin, a firearm was discharged during the commission of the offence;
- in the case of a physical examination application, the offender has distinguishing marks or injuries – these do not have to have been acquired during the commission of the offence;
- there are reasonable grounds to believe that the conduct of the procedure on the person may tend to confirm or disprove his or her involvement in the commission of the offence; and
- the person has refused to give their consent to a request to submit to a compulsory procedure or the person is incapable of giving informed consent by reason of mental impairment.²⁰⁷

In addition, the court has to be satisfied that in all the circumstances the making of the order is justified.²⁰⁸

As in applications to the Children's Court for fingerprints, when an application is made to a Magistrates' Court for an order directing a person to undergo a compulsory procedure, the person upon whom the procedure is sought to be undertaken, cannot call or cross-examine witnesses and is confined in terms of the matters in respect of which he or she can address the court.²⁰⁹

A “non-intimate compulsory procedure”, the taking of a “non-intimate sample” or a physical examination of a “non-intimate part of the body”, may be authorised by a senior officer.²¹⁰ A “senior police officer” is defined by [s 464 of the Crimes Act 1958](#) to mean an officer of or above the rank of senior sergeant. A senior police officer, not involved in investigating the offence for which

the compulsory procedure is required, may authorise another police officer to conduct a “non-intimate compulsory procedure” on a relevant suspect where the senior officer is satisfied that:

- the person is not under 18 years or incapable of giving informed consent by reason of mental impairment;
- the person is under lawful arrest or in police custody subject to an order under [s 464B](#);
- the person has refused to consent to the procedure;
- there are reasonable grounds to believe the person has committed the relevant offence;
- the matters that a court must be satisfied of, regarding the location of relevant evidentiary material or discharge of a firearm etc (set out above), are met;²¹¹ and
- the authorisation is justified in all the circumstances.²¹²

A senior officer must not authorise a “non-intimate compulsory procedure” where an application in relation to the same matter on the same grounds has been refused by a court or a senior officer.²¹³

An authorisation in either of these circumstances is void.²¹⁴ The authorisation must be in writing and record the date and time it was given, the grounds on which it was given and the sample or examination authorised.²¹⁵

A typical example of a case in which a physical examination of a suspect is appropriate is where a complainant in a sex case has described particular characteristics of the offender, such as tattoos of a particular kind.²¹⁶

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[ID.3400] Forensic procedures upon children

Significant limitations are imposed by the legislature in relation to the conduct of forensic procedures upon children. Police are precluded specifically from even requesting a child to undergo a forensic procedure or a compulsory procedure if the child is under 10 years of age and is suspected of having done anything, or omitted to do anything, that would have constituted an offence had the child been of the age of criminal responsibility.²¹⁸

Police are not entitled to request a child aged 10 years or more but under 18 years to undergo either kind of procedure even if the child is suspected of having committed, been charged with or summonsed to answer a charge for an offence unless the Children's Court has authorised it.²¹⁹ [Section 464U\(3\)](#) specifies a series of violence, drug and other offences in respect of which application may be made.²²⁰ It is much more limited than the offences in respect of which a comparable application can be made in respect of adults.

The Children's Court can make an order for the child's submission to a compulsory procedure if satisfied, on the balance of probabilities, of a number of matters. The matters specified are almost identical to those applying in relation to court applications for adults. First, the court must be satisfied that the child is suspected or charged with committing the offences listed in s 464U(3),²²¹ and secondly that there are reasonable grounds to believe that the child has committed the offence in respect of which the application is made.²²² Save in an application to take a gunshot residue sample, it must also be satisfied that:

- material reasonably believed to be from the body of a person who committed the offence has been found at the scene of the offence, on the victim, on anything reasonably believed to have been worn or carried by the victim when the offence was committed or on an object or person reasonably believed to have been associated with the commission of the offence; or
- there are reasonable grounds to believe that because of the nature of the offence or injuries inflicted during the offence, material from the body or clothing of the victim is present on the offender or anything reasonably believed to have been worn or carried by the offender when the offence was committed or on an object reasonably believed to have been associated with the commission of the offence.²²³

The court must also be satisfied that:

- in the case of a gunshot residue application for a sample or washing from the skin, a firearm was discharged during the commission of the offence;
- in the case of a physical examination application, the offender has distinguishing marks or injuries – these do not have to have been acquired during the commission of the offence; and
- there are reasonable grounds to believe that the conduct of the procedure on the child may tend to confirm or disprove his or her involvement in the commission of the offence.²²⁴

In addition, the court has to be satisfied that in all the circumstances the making of the order is justified.²²⁵ In this regard, the court is obliged to take into account at least the seriousness of the circumstances surrounding the commission of the offence, the alleged degree of participation by the child in the commission of the offence and the age of the child.²²⁶ As with other applications within this Division, the child has only limited rights of participation, these not including being a party or the right to call or cross-examine witnesses. In addition, the child may not address the court save in relation to the criteria for the granting of the order or in relation to the criteria on the basis of which the making of an order is justified in all the circumstances.²²⁷

[ID.3420] Interim orders

On occasions, there is the potential for samples or evidence to be lost if there is delay occasioned by the need to proceed with curial process in order to procure an order enabling physical examination. For this reason a member of the police can apply for an interim order directing a person to undergo

a compulsory procedure.²²⁸ However, the exception to this is provision of blood samples. Interim orders cannot apply to these.²²⁹ In order to apply for an interim order, the police member must believe on reasonable grounds that the sample or evidence sought to be obtained by the compulsory procedure is likely to be lost or destroyed if the procedure is delayed until the final determination of the application. This forms the essence of what the court needs to be satisfied of, together with the core elements of what otherwise entitles a court to make an order relating to a compulsory procedure under [s 464T\(3\)](#) or [s 464U\(7\)](#) of the *Crimes Act 1958*. The application may be made by telephone if the urgency of the situation makes such a procedure necessary.²³⁰ However, detailed procedures must be followed in accordance with [s 464W](#).

[ID.3440] Warrants

A warrant may be issued by the Magistrates' Court or the Children's Court for a member of the police:

- to break, enter and search any place where the person named is suspected to be;
- to arrest the person;
- to bring them before the court for the hearing of the application; and
- if the application is granted, to detain them for as long as reasonably permits the conduct of the compulsory procedure.²³¹

[ID.3460] Procedure for taking samples

[Section 464Z of the Crimes Act 1958](#) differentiates between intimate and non-intimate samples and the conduct of physical examinations of “intimate” and “non-intimate” parts of the body. Dental samples are also placed in a category of their own. For definitions of “intimate sample”, “intimate part of the body”, “non-intimate sample”, “non-intimate part of the body” and “physical examination”, see [\[ID.3340\]](#). A dental impression is not defined.

Immediately before a forensic procedure is conducted the person on whom the procedure is to be conducted must be informed by a member of the police force that they do not have to answer any questions of the medical practitioner, nurse or other person conducting the procedure but anything they do say may be used in evidence. The giving of this information must be recorded in writing signed by the person or by audio or audiovisual means.²³²

A series of protections are prescribed to apply to the taking of samples. An intimate sample or a physical impression of an intimate part of the body may only be taken or conducted by a doctor or nurse of the same gender as the suspect, if that is practicable.²³³ Not surprisingly, a dental impression can only be taken by a dentist.²³⁴ A non-intimate sample or a physical impression of a non-intimate part of the body may be taken or conducted by a doctor or a nurse or a person who is authorised.²³⁵

A person required to give a mouth scraping may request they provide a self-taken mouth scraping as an alternative and may do so if the authorised person conducting the forensic procedure considers it appropriate in the circumstances. The consent by the person making the request must be recorded in writing or by audio or audio-visual means.²³⁶ A person required to give a hair sample (other than a pubic hair) may request they provide self-taken mouth scraping as an alternative and may do so if the authorised person conducting the forensic procedure considers it appropriate in the circumstances and again the consent by the person making the request must be recorded in writing or by audio or audio-visual means.²³⁷ The person from whom a sample is to be taken may request a doctor, nurse or dentist, as the case may be, of their own choosing to be present, except where they are taking a scraping from their own mouth.²³⁸ Samples taken or physical examinations conducted must be done “in a manner consistent with the appropriate medical or dental standards”,²³⁹ in the presence of a police member if practicable of the same gender as the suspect (unless the sample is a self-taken mouth scraping)²⁴⁰ and in circumstances affording reasonable privacy to the suspect and in the presence of only those persons expressly permitted to be present.²⁴¹ Not more than 10 millilitres of blood may be taken.²⁴² Reasonable force can be used pursuant to a senior police officer's authorisation (non-intimate compulsory procedure only) or a court order by a member of the police force, of the same gender as the person subject to the procedure and not involved in the relevant investigation, to assist a doctor, nurse or dentist or authorised person to conduct the forensic procedure.²⁴³ Other than the taking of blood samples or self-taken mouth scrapings, the taking of an intimate sample or the examination of an intimate part of a person's body by order of a court must be video-recorded if practicable, if the person consents. The procedure must at least be witnessed by an independent doctor, nurse or dentist or such professional chosen by the person.²⁴⁴

The *Victoria Police Manual – Operations* contains specific instructions for police in relation to forensic procedures. It is important to note that the Police Commissioner's Instructions are just that – they are not the law although they are clearly intended to reflect the law in its practical application as far as possible. However, they do provide a sound basis for testing whether proper investigation procedures have been applied in a particular case. In particular the forensic procedure instructions provide useful information on the Victoria Police (VP) forms required for various aspects of the forensic procedure system and basic requirements for the collection, treatment, retention, transport and analysis of crime scene and reference samples.²⁴⁵

[ID.3480] Analysis of samples

Samples taken must be analysed in accordance with the “prescribed standards” if any and “by an authorised analyst”.²⁴⁶

[ID.3500] Analysis of scene of crime materials

When material reasonably believed to be from the body of a person who committed an indictable offence has been found at the scene of the offence, or on the victim, or on anything reasonably believed to have been worn or carried by the victim at the time of the offence, or on an object or person reasonably believed to have been associated with the commission of the offence, the person has rights in respect of further analysis. So too does a suspect when material reasonably believed to be from the body of a victim of an indictable offence which has not been located has been found on a person reasonably believed to have been associated with the commission of the offence. When there is sufficient material to be analysed both in the investigation of the offence and on behalf of the person, a part of the material sufficient for the further analysis must upon request be delivered to that person.²⁴⁷ Similarly, where a sample has been taken from a child in relation to investigation of a sex offence and the child was allegedly conceived as a result of the offence, a person suspected of committing the offence and from whom a sample has been taken in respect of the offence may request, and is entitled to, part of the child's sample, provided there is sufficient material.²⁴⁸

[ID.3520] Expert reports

When a forensic procedure is undertaken, the person from whom a sample has been taken or upon whom a physical examination has been undertaken is entitled to a “forensic report”. It must be given or sent by registered post to the person as soon as is practicable.²⁴⁹

[ID.3540] Evidence relating to forensic procedures

Important provisions deal with the admissibility of evidence obtained as a result of forensic procedures and samples obtained voluntarily.²⁵⁰ In general, infraction of the obligations placed upon police by the Act renders the evidence inadmissible against the person. However, a court is nonetheless entitled to admit the evidence if the prosecution satisfies it on the balance of probabilities that the circumstances justify it or the accused consents to the reception of the evidence.²⁵¹ In making its decision in respect of the justification for the reception of the evidence, the court is stated to be entitled to have regard to:²⁵²

- the probative value of the evidence, including whether equivalent evidence or evidence of equivalent probative value could have been obtained by other means;
- the reasons given for the failure to comply with the obligation imposed by the Act which has been breached;
- the gravity of the failure and whether it deprived the person of a significant protection;
- whether the failure was intentional or reckless;
- the nature of the offence alleged against the person and subject-matter of the proceedings;

- whether reception of the evidence would seriously undermine the protection given to persons under the subdivision of the Act; and
- any other matters the court considers relevant.

It is specifically provided that the probative value of the evidence does not by itself justify the reception of the evidence.²⁵³

The court is required to balance the utility of the proposed evidence with the significance of the breach of the policy objectives of the legislation in dealing fairly with suspected persons, protecting their dignity and their privacy and ensuring that they have an opportunity, so far as possible, to test the same material as that upon which the prosecution proposes to rely. For the most part, the prosecution will be seeking to rely upon the forensic procedure evidence in serious cases of violence. This in itself has the potential to tip the balance in favour of admission. Moreover, in such cases it will be common for the evidence to be of real consequence for the ultimate resolution of the facts in dispute by the jury. This too may entail a significant temptation for investigators to take short cuts or to disadvantage the interests of the accused. The way in which the court should undertake the balancing exercise is not informed by the wording of s 464ZE. However, it can confidently be predicted that where the infraction is technical and unintentional, not seriously disadvantaging the accused, but the evidence is of significance to the proof of the prosecution case and it is a serious offence with which the accused is charged, the evidence will generally be admitted. Similarly, where the infraction is serious and deliberate, resulting in a grave disadvantage to the forensic interests of the accused, it is likely that the court will hold the evidence to be inadmissible.

[ID.3560] Procedures subsequent to the commission of a forensic sample offence

A “forensic sample offence” is any offence specified in *Sch 8 of the Crimes Act 1958*. In addition, see drug trafficking;²⁵⁴ cultivating of a commercial quantity of a narcotic plant;²⁵⁵ cultivation for the purposes of trafficking.²⁵⁶

Where a court finds a person guilty of a “forensic sample offence” or an offence of conspiracy to commit or attempting to commit a forensic sample offence, a member of the police force within six months of the expiration of any appeal period, or final determination of an appeal, may apply to a court for an order directing the person to undergo a forensic procedure for the taking of a sample from any part of the body. A court may make such an order²⁵⁷ and can make consequential orders permitting breaking, entering, searching, arresting, conducting a person to court and detention of a person in custody to enable the carrying out of such orders.²⁵⁸ The legislation also provides for “retrospective applications”. If a person has been found guilty of a forensic sample offence before s 25 of the *Crimes (Amendment) Act 1997*, and is serving a period of imprisonment or detention in a prison, police gaol, youth justice centre or as a security patient for any offence a member of

the police force may apply to the Magistrates' Court or the Children's Court, as appropriate, for an order directing the person to undergo a forensic procedure for the taking of a sample from any part of the person's body.²⁵⁹ In doing so, the police member must specify whether it is an intimate or non-intimate sample for which they are applying.²⁶⁰ A court may make such an order but in determining whether to do so must take into account the seriousness of the circumstances of the forensic sample offence, must be satisfied that, in all the circumstances, the making of the order is justified, and may make such inquiries on oath or otherwise as it considers desirable.²⁶¹ A finding of guilt is not, of itself, sufficient to justify the making of the order. Without more such an order is unlikely to be made.²⁶²

Gillard J has held that the making of an order by a magistrate that a person undergo a forensic procedure prejudices a person's rights in that it legitimises what would otherwise constitute a criminal battery and a civil assault. Accordingly, he has held that the magistrate's decision-making process must be conducted fairly.²⁶³ He found this to require that the magistrates be:²⁶⁴

[U]nbiased, that he or she require adequate evidence of the matters that have to be considered and that the Magistrate comply with the statutory requirements, namely, the pre-requisites found in s 464ZF(3)(b) and a consideration and determination of the matters in s 464ZF(8) and any other relevant matter before an order is made. It is to be noted that the Magistrate is obliged to give reasons for the decision and to cause a copy of the order and reasons to be served on the person affected by the order.

He held that:²⁶⁵

[T]he first matter for the court to consider is the seriousness of the circumstances of the forensic sample offence. It would be relevant to consider the prisoner's criminal record and the propensity if any of further offending. In addition, the magistrate should take into account that the scope and purpose of the section is to obtain DNA samples not only to assist in the investigation of past offences but also to provide some deterrence to the person providing the DNA sample from committing offences in the future. In addition, the sample may be used to investigate future offences. It is factors such as those which a magistrate should take into account in determining whether the order should be made. Overall, in considering all matters the Magistrate must be satisfied that in all the circumstances the making of the order is justified.

Gillard J found that the legislative intention was to exclude the obligation to give notice to an adult and an opportunity to be heard.²⁶⁶ From this he found it followed that, in respect of adults, there was no entitlement to be notified of the hearing. In short, he determined that:²⁶⁷

The minor nature of the infringement of rights, the small impact of execution of the order on the person concerned, the legislative purpose and the fact that the decision was left to a responsible and careful judicial officer all support my conclusion that the procedural fairness rule in the circumstances of this case do not require notice to the applicant and a right to be heard prior to the order being made.

However, he found that applications for such orders should be made in open court and that proper reasons should be provided by magistrates in respect of such orders:²⁶⁸

[T]he reasons should as a minimum state that the seriousness of the circumstances of forensic sample offence have been considered, the effect of the consideration and that the said circumstances if it be the case lead to the conclusion that prima facie an order should be made and the reasons why. Further, the reasons should state that there is a reason or there are reasons why the order is justified and those reasons should be given.

Finally, he held that the requirement for service upon the prisoner of the reasons and order:²⁶⁹

[M]eans personal service on the person who should be given the opportunity to read and understand the court order prior to the procedure being carried out.

Where a forensic procedure is conducted on a person and a court finds a person guilty of the offence in respect of which the forensic procedure was conducted, any other offence arising out of the same circumstances or any other offence in respect of which evidence obtained as a result of the forensic procedure had probative value, an application can be made to a court for an order permitting the retention of the sample taken and any related material and information.²⁷⁰ Such an application can be made by a member of the police force within six months after the expiry of the appeal period or after the final determination of the appeal. In deciding whether to make such an order a court is obliged to take into account the seriousness of the circumstances of the offence, must be satisfied that in all the circumstances the making of the order is justified and may make such inquiries on oath or otherwise that it considers desirable. These requirements emphasise the breadth of evaluative assessment required of the court before it can conclude the retention order is justified. The court is required to identify all the circumstances that militate against as well those that justify the making of an order.²⁷¹

A key aspect of the legislative regime for the taking of intimate and non-intimate body samples is the procedure for destruction of samples in certain circumstances. Where police make an application for retention of information following a finding of guilt,²⁷² or a court refuses such an application, the Chief Commissioner is obliged without delay to destroy or cause to be destroyed

any sample taken, as well as “related material and information”.²⁷³ Section 464ZFC(3) and (4) of the *Crimes Act 1958* creates criminal offences punishable on conviction by level 8 imprisonment (up to one year) and fine (up to 120 penalty units) for knowing failure to destroy, or the use of such material or information derived from it.

A system of computerised databases is specifically provided for under s 464ZFD. Information obtained from the analysis of samples taken or procedures conducted in accordance with the legislation, and permitted to be retained, “may be included in a computerised database”. This is so even though the information may identify the person.²⁷⁴ Similarly, information obtained from the analysis of samples or procedures conducted in accordance with the legislation and permitted to be retained can be included in a computerised database for statistical purposes, provided that it cannot identify the person.²⁷⁵ The Chief Commissioner has strict reporting obligations to the Attorney-General about samples taken and destruction of materials.²⁷⁶

Specific destruction obligations apply in relation to material and information:²⁷⁷

- where a person has not been charged with a “relevant offence” within 12 months;
- where the person has been so charged but the person is not found guilty of the offence or any other relevant offence before the end of the period.

A “relevant offence” is very broadly defined as:²⁷⁸

- the offence in respect of which the forensic procedure was conducted; or
- any other offence arising out of the same circumstances; or
- any other offence in respect of which the evidence obtained as a result of the forensic procedure has probative value.

However, police can apply to the Magistrates' Court or Children's Court as appropriate for an order extending the period within which they must destroy the material.²⁷⁹ Criteria for making such an order are not enunciated by the Act but where such an order is made reasons must be given by the court.²⁸⁰

Where a child undergoes a forensic procedure and the sample is not otherwise required to be destroyed when the person reaches the age of 26 without being found guilty of any further offence the Chief Commissioner is obliged to destroy or cause the sample to be destroyed.²⁸¹ However, this obligation does not apply in relation to certain serious offences namely, murder, attempted murder, defensive homicide, manslaughter, offences or attempted offences against s 16, 17, 18, 25, 26, 29 or 31, Subdivs (8A), (8B), (8C), (8D), (8E) of Div 1 of Pt I, or against s 75, 75A, 197A or 197 where it is charged as arson.

[ID.3580] Voluntary samples

Persons of or above 18 years of age may voluntarily give samples, be they intimate or non-intimate, if they have received due information and warnings.²⁸² Those who have given their samples in such circumstances may, after giving the sample, withdraw their consent to the retention of the sample.²⁸³ In general this means that their sample must be destroyed²⁸⁴ unless a Magistrates' Court orders otherwise under s 464ZGF of the *Crimes Act 1958* if it is satisfied on the balance of probabilities that there are reasonable grounds to believe that the person has committed the offence in respect of which the application for retention of the sample has been made and that in all the circumstances the making of the order for retention is justified.²⁸⁵

[ID.3600] Forensic procedure after commission of a forensic sample offence

A “forensic sample offence” is defined by Sch 8 of the *Crimes Act 1958*. If, after s 25 of the *Crimes (Amendment) Act 1997* came into effect, a court finds a person guilty of a forensic sample offence or an offence of conspiracy to commit, incitement to commit or attempting to commit a forensic sample offence, an application may be made to a court for an order directing the person to undergo a forensic procedure.²⁸⁶ In addition, if after s 25 of the *Crimes (Amendment) Act 1997* came into force, a person has been found guilty of a forensic sample offence and at any time is serving a sentence of imprisonment of a period of detention in a prison, police gaol, or youth training centre, or as a security patient in a mental health service for any offence, a member of the police force may apply to a Magistrates' Court or Children's Court, as appropriate, for an order directing the person to undergo a forensic procedure for the taking of a sample from any part of the body. The court may make an order accordingly.²⁸⁷

In determining either such application, a court must take into account the seriousness of the circumstances of the forensic sample offence, must be satisfied that, in all the circumstances the making of the order is justified, and may make such inquiries on oath or otherwise as it considers justifiable.²⁸⁸ It can make consequential orders to enforce such orders.²⁸⁹

Footnotes

- 1 See CJ Lennard and T Paterson, “Dactyloscopy: The science of fingerprinting” in I Freckelton and H Selby (eds), *Expert Evidence* (Thomson Reuters subscription service, 1993–), Ch 96; CJ Lennard and T Paterson, “Dactyloscopy: The science of fingerprinting” in I Freckelton and H Selby (ed), *Expert Evidence in Criminal Law* (LBC, 1999), Ch 3.
- 2 See CJ Lennard and T Paterson, “Dactyloscopy: The science of fingerprinting” in I Freckelton and H Selby (eds), *Expert Evidence* (Thomson Reuters subscription

service, 1993–), Ch 96; CJ Lennard and T Paterson, “Dactyloscopy: The science of fingerprinting” in I Freckelton and H Selby (ed), *Expert Evidence in Criminal Law* (LBC, 1999), Ch 3.

R v Krausch (1913) 32 NZLR 1229 disagreeing with *R v Parker* [1912] VLR 152.

R v Carr [1972] 1 NSWLR 608 at 611–612.

See *R v Buisson* [1990] 2 NZLR 543 at 549.

See *R v Lawless* [1974] VR 398 at 423; *R v O'Callaghan* [1976] VR 676 at 678.

R v Buisson [1990] 2 NZLR 543.

R v Buisson [1990] 2 NZLR 543 at 546.

D Campbell, “Fingerprints: A review” (1985) Criminal LR 195 at 199.

R v Nielsen (1984) 16 CCC (3d) 39 at 70.

See *Victoria Police Manual*, VPM 116–6 Aids to Investigation – Fingerprinting.

See *Grollo v Bates* (1994) 53 FCR 218; 75 A Crim R 271; 125 ALR 492 at 495 (ALR), perinfeld J; see also *R v Ireland* (1970) 126 CLR 321; 44 ALJR 263.

Crimes Act 1958, s 464L(1).

Crimes Act 1958, s 464L(1)(a).

Pursuant to *Crimes Act 1958*, s 464M(5).

Crimes Act 1958, s 464L(3)(a)–(f).

Crimes Act 1958, s 464L(4).

Crimes Act 1958, s 464L(6).

Crimes Act 1958, s 464M(5).

Crimes Act 1958, s 464M(6).

Crimes Act 1958, s 464M(7).

Crimes Act 1958, s 464M(13)(a).

Crimes Act 1958, s 464M(13)(b).

Crimes Act 1958, s 464M(9)(b).

Crimes Act 1958, s 464M(12)(a)–(c).

Crimes Act 1958, s 464K(1). See also *Watkins v Victoria* (2010) 27 VR 543; 203 A Crim R 20; [2010] VSCA 138.

Crimes Act 1958, s 464K(2).

See below at [ID.3100] under heading “Destruction of fingerprinting records”.

Crimes Act 1958, s 464K(7).

Crimes Act 1958, s 464N.

Crimes Act 1958, s 464N.

Crimes Act 1958, s 464O(2)–(3).

Crimes Act 1958, s 464O(4).

Crimes Act 1958, s 464O(5).

Crimes Act 1958, s 464P(1).

Crimes Act 1958, s 464P(2).

Crimes Act 1958, s 464Q(1).

38 *Crimes Act 1958*, s 464Q(2)(a)–(b).

39 *Crimes Act 1958*, s 464Q(3).

40 *Crimes Act 1958*, s 464NA(1).

41 *Crimes Act 1958*, s 464NA(6).

42 *Crimes Act 1958*, s 464NA(3).

43 *Crimes Act 1958*, s 464NA(4).

44 *Crimes Act 1958*, s 464NA(5).

45 *Crimes Act 1958*, s 464NA(7).

46 The author acknowledges the assistance of Dr Dadna Hartman, Manager Molecular Biology, Victorian Institute of Forensic Medicine and Dr Henry Roberts, Forensic Scientist, Leader of DNA Interpretation and Statistics Unit and Dr Rebecca Heyes, Acting Assistant Director, Biology Division, Victoria Police Forensic Services Department, each of whom offered comments on the draft of this section. For an extensive examination of the techniques of DNA profiling, see S Gutowski, “DNA typing in criminal investigations” in I Freckelton and H Selby (ed), *Expert Evidence* (Thomson Reuters, subscription service, 1993–), Ch 80; S Gutowski, “DNA typing evidence in criminal investigations” in I Freckelton and H Selby (ed), *Expert evidence in criminal law* (LBC, 1999), pp 4–45. See also C Pearman, “Parentage testing” in I Freckelton and H Selby (ed), *Expert Evidence*, (Thomson Reuters, subscription service, 1993–), Ch 80; Pearman, “Parentage testing” in I Freckelton and H Selby (ed), *Expert Evidence in Family Law* (LBC, 1999), pp 745–803.

47 X chromosome DNA is less variable than the STR regions of the autosomes as there is less possibility for exchange of DNA between sex chromosomes inherited from each parent (males do not inherit an X chromosome from their father). It remains possible a role will be identified for X chromosome DNA in criminal investigations but at the time of writing this was not the case.

48 Gametes contain only half of the DNA information that is present in other cell types.

49 *R v Jarrett* (1994) 62 SASR 443; 73 A Crim R 160 at 163 (A Crim R).

50 S Gutowski, “DNA typing in criminal investigations” in I Freckelton and H Selby (ed), *Expert Evidence* (Thomson Reuters, subscription service, 1993–) at [80.160]; S Gutowski, “DNA typing evidence in criminal investigations” in I Freckelton and H Selby (ed), *Expert Evidence in Criminal Law*, (LBC, 1999), pp 12–13.

51 *R v Karger* (2001) 83 SASR 1; [2001] SASC 64 at [103]–[104].

52 See National Research Council, *DNA technology in forensic science* (National Academy Press, 1992); National Research Council, *The evaluation of forensic DNA evidence* (National Academy Press, 1996). See further I Freckelton, “DNA profiling evidence: Legal issues” in I Freckelton and H Selby (ed), *Expert*

Evidence (Thomson Reuters, subscription service, 1993–), Ch 13; I Freckelton, “DNA profiling evidence” in I Freckelton and H Selby, *The Law of Expert Evidence* (LBC, 1999), pp 436–503.

53 [R v McNeill \(No 2\) \[2007\] NFSC 3](#) at [29].

54 See [R v Tuite \[2014\] VSC 662](#) at [20]–[29] and [Tuite v The Queen \[2015\] VSCA 148](#) at [22]–[32] and [119]–[120] for a discussion of the operational mechanics of STRmix software. Tuite is a case where both P+ and PP21 analysis were used on different samples. See discussion at [\[ID.3260\]](#) “Presenting the significance of the results”.

55 The kit currently used is Life Technologies’ “Yfiler” kit.

56 A picogram is 0.001 of a nanogram and a nanogram is one billionth of a gram.

57 B Budowle, AJ Eisenberg, A van Daal, “Validity of low copy number typing and applications to forensic science” (2009) 50(3) *Croat Med J* 207–217 available from: <http://www.cmj.hr/>. The understanding of DNA science, and associated scientific techniques, are constantly improving. It is always important to consider whether what was not possible yesterday is possible today.

58 For a discussion of these complex issues see B Budowle, “Low copy number typing still lacks robustness and reliability” *Internet* [2010] [cited: 25 Sep 2012]. Available from: <http://www.promega.com/resources/articles/profiles-in-dna/low-copy-number-typing-still-lacks-robustness-and-reliability/>.

59 Named after Edmond Locard, early 20th century French medical examiner and pioneer forensic scientist. Author of the seven volume *Traite de criminalistique* among other forensic science publications.

60 For a detailed analysis of the development of Australian case law see I Freckelton, “DNA profiling evidence: Legal issues” in I Freckelton and H Selby (ed), *Expert Evidence* (Thomson Reuters subscription service, 1993–) at [14.610]–[14.710].

61 *Evidence Act 2008*, ss 76, 79.

62 [R v Bonython](#) (1984) 38 SASR 45 at 46–47, per King CJ.

63 [Dasreef Pty Ltd v Hawchar](#) (2011) 243 CLR 588; 85 ALJR 694; 9 DDCR 25; 277 ALR 611; [2011] HCA 21 at [37], per French CJ, Gummow, Hayne, Crennan, Kiefel and Bell JJ.

64 *Evidence Act 2008*, s 55(1).

65 *Frye v United States* 293 F 1013 (1923) at 1014, quoted with approval by Mullighan J in [R v Karger](#) (2001) 83 SASR 1; [2001] SASC 64 at [171].

66 [R v Karger](#) (2001) 83 SASR 1; [2001] SASC 64 at [172]. In essence this is the process Mullighan J undertook on the voir dire in *R v Karger* when examining the developments in DNA technology including Profiler Plus system, the Short Tandem Repeat (STR) analysis, fluorescence technology, Genescan analysis software and multiplex technology.

- 67 *Daubert v Merrell Dow Pharmaceuticals, Inc* (1993) 509 US 579 at 590 per Blackmun J
- 68 *R v Tang* [2006] NSWCCA 167; (2006) 65 NSWLR 681; 161 A Crim R 377 at [137]–[139]. This approach is subject to some strong criticism, see “Edmond G, “The admissibility of forensic science and medicine evidence under the Uniform Evidence Law” (2014) 38 Crim LJ 136. See also *Tuite v The Queen* [2015] VSCA 148 at [51]–[71], where the Court of Appeal considered and concluded the approach in *Tang* was correct and the commentary at [EA.79.40] “Specialised knowledge” and [EA.137.40] “Reliability of evidence is relevant to discretion”. See Edmond G, “The admissibility of forensic science and medicine evidence under the Uniform Evidence Law” (2014) 38 Crim LJ 136, Table 1 at p 144.
- 69 *Honeysett v The Queen* (2014) 311 ALR 320; (2014) 88 ALJR 786; [2014] HCA 29. See also *Tuite v The Queen* [2015] VSCA 148 at [51]–[71].
- 70 *Honeysett v The Queen* (2014) 311 ALR 320; (2014) 88 ALJR 786; [2014] HCA 29 at [42] – [43].
- 71 Edmond G “The admissibility of forensic science and medicine evidence under the Uniform Evidence Law” (2014) 38 Crim LJ 136 at pg 150. The reference is to *Dasreef Pty Ltd v Hawchar* (2011) 243 CLR 588; (2011) 277 ALR 611; (2011) 85 ALJR 694; [2011] HCA 21. See also discussion of the *Victorian Supreme Court Practice Direction No 2 of 2014 – Expert Evidence in Criminal Trials* at [ID.2500] “The opinion rule – specialised knowledge and expert evidence of identification”.
- 72 See for example *R v Button* [2001] QCA 133. The Innocence Project in the United States records 325 wrongful convictions in the United States overturned by post-conviction DNA evidence as at 15 February 2015, see <http://www.innocenceproject.org/>.
- 73 *Hillier v The Queen* (2008) 163 ACTR 60; 163 ACTR 60; [2008] ACTCA 3. See particularly at [49]–[119] and [166]–[171]. The appeal was re-heard by the Australian Capital Territory Supreme Court following a successful appeal by the prosecution to the High Court, *R v Hillier* (2007) 228 CLR 618; 81 ALJR 886; 233 ALR 634; [2007] HCA 13. During the re-hearing of the appeal in the Supreme Court further evidence, both written and oral, was received from three expert DNA witnesses.
- 74 *Hillier v The Queen* (2008) 163 ACTR 60; 163 ACTR 60; [2008] ACTCA 3 at [181], per curium.
- 75 Hon FHR Vincent, *Report – Inquiry into the circumstances that led to the conviction of Mr Farah Abdulkadir Jama* (Victorian Government Printer, May 2010).
- 76 Hon FHR Vincent, *Report – Inquiry into the circumstances that led to the conviction of Mr Farah Abdulkadir Jama* (Victorian Government Printer, May 2010).
- 77

2010), p 29 quoting from the report of the Royal Commissioner conducting the 1984 South Australian inquiry into the case of Edward Charles Splatt.

78 *Evidence Act 2008*, ss 76, 79.

79 *R v Jarrett* (1994) 62 SASR 443; 73 A Crim R 160 at 171 (A Crim R), per Mullighan J. See also *Chamberlain v The Queen (No 2)* (1984) 153 CLR 521; 58 ALJR 133; 51 ALR 225; *R v Humphrey* (1999) 72 SASR 558; 103 A Crim R 434; [1999] SASC 67.

80 See, in particular, *Gordon v The Queen* [1995] 1 Cr App R 290 at 296; see also *Adams v The Queen* [1996] 2 Cr App R 467; *Doheny v The Queen* [1997] 1 Cr App R 369. See also for Scotland *Welsh v HM Advocate* [1992] SLT 193.

81 *R v Johnston* (1992) 69 CCC 395; *R v Baptiste* (1994) 88 CCC (3d) 211; *R v Terreira* (1998) 123 CCC (3d) 1.

82 *R v Pengally* (1991) 7 CRNZ 333; *R v Dougherty* (1996) 14 CRNZ 145.

83 For an early case, see *People v Castro* 545 NYS 2d 985 (1989); see later *United States v Bonds* 12 F 2d 540 (1994); *Caldwell v State* 393 SE 2d 436 (1990); *People v Miles* 577 NE 2d 477 (1991); *People v Axell* 235 Cal App 3d 836 (1991).

84 For an extensive analysis of the decisions internationally that have examined the admissibility of DNA profiling evidence, see I Freckelton, “DNA profiling evidence: Legal issues” in I Freckelton and H Selby (eds), *Expert Evidence* (Thomson Reuters subscription service, 1993–), Ch 14; I Freckelton, “DNA profiling evidence” in I Freckelton and H Selby (eds), *The Law of Expert Evidence* (Thomson Reuters, 1999), pp 436–503.

85 *R v Pantoja* (1996) 88 A Crim R 554 at 558, per Hunt CJ at CL. In doing so, the court applied the test enunciated in *Frye v United States* 293 F 1013 (1923) as applied in *R v Gilmore* [1977] 2 NSWLR 935 at 939–941. For an analysis of this issue, see I Freckelton, “The area of expertise rule” in I Freckelton and H Selby, *The Law of Expert Evidence* (Thomson Reuters, 1999), pp 48 ff; I Freckelton, “The area of expertise rule” in I Freckelton and H Selby (ed), *Expert Evidence* (Thomson Reuters subscription service, 1993–), Ch 9.

86 *Director of Public Prosecutions (Vic) v Paulino* (2017) 54 VR 109; [2017] VSCA 38 provides an example of the care and rigour required when assessing relevance of DNA evidence.

87 *Makita (Aust) Pty Ltd v Sprowles* (2001) 52 NSWLR 705; 25 NSWCCR 218; [2001] NSWCA 305.

88 *R v Sing* (2002) 54 NSWLR 31; [2002] NSWCCA 20 at [35], per Hodgson JA, Levine and Howie JJ agreeing. See also *R v Ryan* [2002] VSCA 176 where the matter was further complicated when the blood sample and the database records relating to the appellant's DNA typing were destroyed whilst the appeal was pending. Compare *Sharwood v The Queen* [2006] NSWCCA 157 where the admissibility of the scientific evidence was conceded at trial.

- 89 For a comprehensive discussion of DNA profiling evidence, its uses, limitations and the development of the case law in Australia, Britain, Canada and the United States, see I Freckelton, “DNA profiling evidence: Legal issues” in I Freckelton and H Selby (eds), *Expert Evidence* (Thomson Reuters subscription service, 1993–), Ch 14.
- 90 See [EA.55.20]–[EA.55.60] of this service and Odgers Uniform Evidence Law [EA.55.60] ff for commentary on s 55 of the *Evidence Act 2008*. *Director of Public Prosecutions (Vic) v Paulino* (2017) 54 VR 109; [2017] VSCA 38.
- 91 See [EA.PT3.11.20] of this service and Odgers Uniform Evidence Law [EA.ch.3.Pt.3.11.30] ff for commentary on Part 3.11 of the *Evidence Act 2008*.
- 92 *R v GK* (2001) 53 NSWLR 317; 125 A Crim R 315; [2001] NSWCCA 413 at [29], per Mason P quoting from King CJ in *Duke v The Queen* (1979) 22 SASR 46; 1 A Crim R 39 at 48 (SASR); at 41 (A Crim R).
- 93 *Evidence Act 2008*, s 137. Relevant DNA cases include *Director of Public Prosecutions (Vic) v Paulino* (2017) 54 VR 109; [2017] VSCA 38; *Ramaros v The Queen* [2018] VSCA 143; *Farha v The Queen* [2018] VSCA 310.
- 94 *R v Jarrett* (1994) 62 SASR 443; 73 A Crim R 160 at 175 (A Crim R), per Mullighan J. See too *R v Karger* (2001) 83 SASR 1; [2001] SASC 64.
- 95 *R v Jarrett* (1994) 62 SASR 443; 73 A Crim R 160 at 169, 171, 173 (A Crim R).
- 96 *R v Jarrett* (1994) 62 SASR 443; 73 A Crim R 160 at 169, 172–173 (A Crim R) and cf with *R v McNeill (No 2)* [2007] NFSC 3 at [46]–[50], per Weinberg CJ.
- 97 *R v Percerep* [1993] 2 VR 109; (1993) 65 A Crim R 419.
- 98 *R v Van Hung Tran* (1990) 50 A Crim R 233 at 242, per McInerney J. See also *R v McNeill (No 2)* [2007] NFSC 3.
- 99 *R v Lucas* [1992] 2 VR 109; (1992) 55 A Crim R 361.
- 100 *R v Lucas* [1992] 2 VR 109; (1992) 55 A Crim R 361 at 113 (VR); at 366 (A Crim R), per Hampel J.
- 101 *R v Lucas* [1992] 2 VR 109; (1992) 55 A Crim R 361 at 118 (VR); at 370 (A Crim R).
- 102 It is to be noted both Hampel J (*R v Lucas* [1992] 2 VR 109; (1992) 55 A Crim R 361 at 116 (VR); at 368 (A Crim R)) and Mullighan J (*R v Jarrett* (1994) 62 SASR 443; 73 A Crim R 160 at 170 (A Crim R)) quote the same passage from Jenkinson J in the Federal Court in *Chamberlain v The Queen (No 2)* (1983) 72 FLR 1; 46 ALR 493 dealing with the “foetal haemoglobin” evidence. In each case Hampel J and Mullighan J quote Gibb CJ and Mason J in *Chamberlain v The Queen (No 2)* (1984) 153 CLR 521; 58 ALJR 133; 51 ALR 225 at 558 (CLR) where they quote Jenkinson J regarding the role of the jury in evaluating conflicting expert evidence.
- 103 *R v Humphrey* (1999) 72 SASR 558; 103 A Crim R 434; [1999] SASC 67.

- 104 *R v Humphrey* (1999) 72 SASR 558; 103 A Crim R 434; [1999] SASC 67 at [19]–[24].
- 105 *R v Humphrey* (1999) 72 SASR 558; 103 A Crim R 434; [1999] SASC 67 at [41]–[44].
- 106 *R v Juric* [2003] VSC 382 at [70] - [71]. Nettle J undertakes a helpful discussion of prejudicial effect and fairness, including reference to some significant cases, at [43] – [69]. The reference to the decision of the Court of Appeal is a reference to *R v Juric* (2002) 4 VR 411; 129 A Crim R 408; [2002] VSCA 77 and see discussion of that decision below. See also *R v ST* [2022] VSC 450 where Taylor J dealt with evidence of likelihood ratios from two different Australian ethnic databases. In *ST* result of the Australian Caucasian Database was allowed to be admitted to show some of the ten accused could be excluded from the DNA found on the knife alleged to be the weapon but the likelihood ratio related to that database was not admitted. The likelihood ratio obtained by applying the Australian Asian Database to the same DNA analysis results was admitted into evidence where the likelihood ratios for one accused (*ST*) was 8 to 1 and another accused (*AK*) it was 9 to 1, see [36] - [37] and [108]. As with the reasons of Nettle J, Taylor J undertakes a helpful discussion of the principles and cases in the context of the evidence before him.
- 107 *R v Juric* (2002) 4 VR 411; 129 A Crim R 408; [2002] VSCA 77.
- 108 *R v Juric* (2002) 4 VR 411; 129 A Crim R 408; [2002] VSCA 77 at [18], per curiam. It should be noted that the Profiler Plus system was new at the time of testing and was “still being explored” by the scientist. This is no longer the case and the Profiler Plus system is widely accepted by the courts.
- 109 *R v Joyce* (2002) 173 FLR 322; 136 A Crim R 465; [2002] NTSC 70.
- 110 *R v Joyce* (2002) 173 FLR 322; 136 A Crim R 465; [2002] NTSC 70 at [8]–[9].
- 111 *Director of Public Prosecutions (Vic) v Wise* [2016] VSCA 173.
- 112 *Director of Public Prosecutions (Vic) v Wise* [2016] VSCA 173 at [70] per curiam, footnotes omitted.
- 113 *R v McNeill (No 2)* [2007] NFSC 3. The Uniform Evidence Act applied to *R v McNeill (No 2)* as it was conducted under the *Evidence Act 2004* (NI).
- 114 *R v McNeill (No 2)* [2007] NFSC 3 at [46]–[50], per Weinberg CJ. The expert witness had examined a total of five hair samples using mitochondrial DNA. Weinberg CJ held the evidence was relevant and the expert’s lack of familiarity with the ethnic make up of Norfolk Island was a matter that went to the weight of her evidence. The witness would not go further than to say she could not exclude the possibility that three of the hairs came from the victim or her maternal relatives. When pressed on the voir dire to give an indication of the frequency she said she could not give a number, but said she would not expect to see the particular profile on Norfolk Island in high frequency.

- 115 *R v Noll* [1999] 3 VR 708; [1999] VSCA 164 at [19]–[20], per Callaway JA.
- 116 For example see *R v Rye* (2007) 178 A Crim R 345; [2007] VSCA 247 at [49]–[51], [57], per Kellam JA, Vincent JA and Whelan AJA agreeing. Compare *R v McNeill (No 2)* [2007] NFSC 3, where the evidence was excluded. Coincidentally both cases involved hair and mtDNA.
- 117 *R v Stokes* [2000] NTSC 12.
- 118 For example see *Director of Public Prosecutions v Tuite* [2014] VSC 662 at [102] – [108] per Emerton J.
- 119 *R v Berry*; *R v Wenitong* (2007) 17 VR 153; 176 A Crim R 195; [2007] VSCA 202 at [35]–[36], per Redlich JA. Redlich JA also observed at [34] that no principle or authority had been cited in support of the appellant's assertion that as a matter of public policy DNA evidence should not be admitted once evidence of possible contamination emerged. See also *Sharwood v The Queen* [2006] NSWCCA 157 at [31]; *Allan v The Queen* [2017] NSWCCA 6 at [54]–[66] per Harrison J. Cf *R v Joyce* (2002) 173 FLR 322; 136 A Crim R 465; [2002] NTSC 70.
- 120 Hon FHR Vincent, *Report – Inquiry into the circumstances that led to the conviction of Mr Farah Abdulkadir Jama* (Victorian Government Printer, May 2010), p 11.
- 121 Hon FHR Vincent, *Report – Inquiry into the circumstances that led to the conviction of Mr Farah Abdulkadir Jama* (Victorian Government Printer, May 2010), pp 13, 22–24, 28.
- 122 Hon FHR Vincent, *Report – Inquiry into the circumstances that led to the conviction of Mr Farah Abdulkadir Jama* (Victorian Government Printer, May 2010), p 16.
- 123 Hon FHR Vincent, *Report – Inquiry into the circumstances that led to the conviction of Mr Farah Abdulkadir Jama* (Victorian Government Printer, May 2010), pp 33–35.
- 124 *R v GK* (2001) 53 NSWLR 317; 125 A Crim R 315; [2001] NSWCCA 413 at [96], per Sully J. In *R v GK Evidence Act 1995* (NSW), ss 135 and 137 are examined in detail.
- 125 Attachments A and B to Hon FHR Vincent, *Report – Inquiry into the circumstances that led to the conviction of Mr Farah Abdulkadir Jama* (Victorian Government Printer, May 2010) provide some valuable insight into the nature and detail of the required protocols.
- 126 Some accused persons spend lengthy periods in custody awaiting the result of forensic tests so their trial can proceed. Unnecessary and unjustifiable depletion of forensic resources potentially exacerbates the problem.
- 127 See *R v Smart (No 3)* [2008] VSC 121 and *R v Guingab* [2010] VSC 256 for examples of cases involving issues of this kind.

- 128 [R v GK \(2001\) 53 NSWLR 317; 125 A Crim R 315; \[2001\] NSWCCA 413](#) at [42], per Mason P.
- 129 For a detailed discussion of DNA statistics see Dr H Roberts, “Statistical evaluation of forensic DNA typing” in I Freckelton and H Selby (eds), *Expert Evidence* (Thomson Reuters subscription service, 1993–), Ch 80A. The writer acknowledges the assistance of Dr Roberts when preparing the text in this section. In [R v Pfennig \[2017\] SASCFC 26](#) several different methods of DNA analysis and subsequent calculation of likelihood ratio were used, including analysis by the Netherlands Forensic Institute and the STRmix currently used in Victoria, and the different results obtained are discussed. [Pfennig \[2017\] SASCFC 26](#) was a cold-case where the circumstantial case included evidence alleging the accused’s DNA was on the pyjamas of the child murder victim. On appeal it was noted, at [104], that the earlier less sophisticated analysis was “engulfed and superseded” by the later more discerning methodology and the trial judge’s finding that the various results were proved beyond reasonable doubt was correct, at [111]–[115] per Rule J.
- 130 It takes the name “equation 4.10” from the paragraph number where it is set out in the 1996 report of the National Research Council of the United States National Academy of Sciences *The evaluation of forensic DNA evidence*, National Academy Press, Washington, DC.
- 131 See [Vyater v The Queen \[2020\] VSCA 32](#) at [54] – [57] for a succinct discussion of ‘match probability’.
- 132 It is important to be very careful when choosing the words to express the probability. Confusing “the probability of the match given the proposition is true” with “the probability of the proposition being true” is an example of the “prosecutor’s fallacy”. As to which see discussion following.
- 133 [R v Berry \(2007\) 17 VR 153; 176 A Crim R 195; \[2007\] VSCA 202](#) at [30], per Redlich JA, Buchanan J agreeing and Kellam JA agreeing in a separate judgment at [139]–[142].
- 134 See [Vyater v The Queen \[2020\] VSCA 32](#) at [58] – [86] for a succinct discussion of ‘likelihood ratio’ and its presentation to, and use by, a jury.
- 135 [R v Tuite \[2014\] VSC 662](#) at [19]–[29]. See also [Tuite v The Queen \[2015\] VSCA 148](#) at [22]–[32] and [119]–[120], where the Court of Appeal considered and upheld Emerton J’s decision in [R v Tuite \[2014\] VSC 662](#).
- 136 [R v Tuite \[2014\] VSC 662](#) at [53], [58] and [79], footnotes omitted. To appreciate the reasoning of Emerton J it is necessary to carefully read the decision in full. The articles referred to by Emerton J at [53] are Gill P et al, “DNA Commission of the International Society of Forensic Genetics: Recommendations of the evaluation of STR typing results that may include drop-out and/or drop-in using probabilistic methods”, (2012) 6 Forensic Science International: Genetics 679

and Steele C and Balding D, “Statistical Evaluation of Forensic DNA Profile Evidence”, (2014) 1 Annual Review of Statistics and its Application 361. On appeal in *Tuite v The Queen* [2015] VSCA 148 at [116]–[118], the Court of Appeal quoted Emerton J’s reasons extensively and concluded her Honour’s findings were well open and her Honour had discharged her gatekeeping role with exemplary care and thoroughness. See also discussion of whether inherent reliability should form part of the test under s 79(1) at [ID.3180] “Development of Australian case law – when opinion is admissible as expert testimony”.

137 RMNE is used in Tasmania and some other jurisdictions. An exclusion percentage was the subject of objection in *Aytugrul v The Queen* (2012) 247 CLR 170; 86 ALJR 474; 215 A Crim R 501; 286 ALR 441; [2012] HCA 15. See discussion below.

138 *R v Pantoja* (1996) 88 A Crim R 554 at 558–559.

139 *R v Pantoja* (1996) 88 A Crim R 554 at 562.

140 *R v Pantoja* (1996) 88 A Crim R 554 at 558; see also *Lafferty* (1992) 80 CCC (3d) 150; *R v Baptiste* (1994) 88 CCC (3d) 211.

141 *R v Karger* (2001) 83 SASR 1; [2001] SASC 64.

142 *R v Karger* (2001) 83 SASR 1; [2001] SASC 64 at [660].

143 *R v Milat* (1996) 87 A Crim R 446 at 451–452.

144 *R v Watters* [2000] EWCA Crim 89 at [17], per Kay LJ, In *R v Watters* the DNA match was based on seven different regions of DNA, including the gender indicator. In four matches from cigarette butts found at four different burglaries, involving sophisticated attacks on safes, there were matches at all seven loci and in relation to a fifth burglary a match at five of the seven loci. The forensic DNA expert called by the prosecution conceded in cross examination that if the accused had two brothers the match probabilities would be reduced substantially. The evidence showed the accused had two brothers who had not provided reference samples. The prosecution relied on the DNA evidence, evidence the accused was a male smoker, lived in the locality of the burglaries and that most safe crackers were male. Held on the evidence the matter should not have been left to the jury. Compare *R v Tsekiri* [2017] EWCA Crim 40.

145 A very good example of the reasoning process involved can be found in *R v Carroll* [2010] SASC 156, a trial by judge alone (Sulan J) where the alleged offence was rape.

146 *R v GK* (2001) 53 NSWLR 317; 125 A Crim R 315; [2001] NSWCCA 413 at [40].

147 *Doheny v The Queen* [1997] 1 Cr App R 369 at 373, per Phillips LJ. Part of the passage quoted by Mason P in *R v GK* (2001) 53 NSWLR 317; 125 A Crim R 315; [2001] NSWCCA 413 at [48].

- 148 *Ali v The Queen* [2022] VSCA 31 at [25] per curiam. In *Ali* [2022] VSCA 31, at [42] – [47], the Court of Appeal accepted that both the prosecutor and the trial judge committed the prosecutor's fallacy. The Court concluded no miscarriage of justice arose because the accused implicitly accepted he was a contributor to the DNA samples relied in by the prosecution. The Court observed, [45], the prosecutor's fallacy was not a relevant error unless it was a fact in issue whether the crime scene DNA sample included the accused's DNA. See also discussion at [28] – [31] where the Court of Appeal considered the possibility, identified by the Court in *Doheny v The Queen* [1997] 1 Cr App R 369, where DNA analysis reaches the point where it is possible to construct a unique DNA profile so that the evidence the accused is the source of the crime scene sample is conclusive. The Court of Appeal concluded there was evidence before it that DNA science had developed to that point.
- 149 *R v GK* (2001) 53 NSWLR 317; 125 A Crim R 315; [2001] NSWCCA 413 at [51], per Mason P.
- 150 *Aytugrul v The Queen* (2012) 247 CLR 170; 86 ALJR 474; 215 A Crim R 501; 286 ALR 441; [2012] HCA 15.
- 151 *Aytugrul v The Queen* (2012) 247 CLR 170; 86 ALJR 474; 215 A Crim R 501; 286 ALR 441; [2012] HCA 15 at [23]–[24], per French CJ, Hayne, Crennan and Bell JJ. Heydon J agreeing the outcome and setting out separate reasons.
- 152 *Aytugrul v The Queen* (2012) 247 CLR 170; 86 ALJR 474; 215 A Crim R 501; 286 ALR 441; [2012] HCA 15 at [25]–[31], per French CJ, Hayne, Crennan and Bell JJ. Heydon J agreeing the outcome and setting out separate reasons.
- 153 *Aytugrul v The Queen* (2012) 247 CLR 170; 86 ALJR 474; 215 A Crim R 501; 286 ALR 441; [2012] HCA 15 at [32]–[33], per French CJ, Hayne, Crennan and Bell JJ. Footnotes omitted. See discussion by Heydon J at [68]–[71], of circumstances where expert evidence is called of research into jurors comprehension of technical evidence or placing undue reliance on it.
- 154 *Vyater v The Queen* [2020] VSCA 32 at [73] per curiam.
- 155 *Vyater v The Queen* [2020] VSCA 32 at [75] per curiam.
- 156 *Vyater v The Queen* [2020] VSCA 32 at [35] – [86] per curiam. The Court of Appeal's discussion provides a valuable reflection on the issues DNA evidence can cause for parties, courts and juries. For contrast see *R v Guingab* [2010] VSC 256 where T Forrest J excluded DNA evidence found on rope connected to an alleged murder where the likelihood ratios were in the range of 48 to 1 and other DNA evidence from a thong in the same case was expressed as 110 billion to 1. T Forrest J concluded that in the circumstances of the case before him there was a real risk of unfair prejudice that could not be adequately addressed by directions to the jury. See also *R v ST* [2022] VSC 450 at [65] – [per Taylor J.

- 157 [R v GK \(2001\) 53 NSWLR 317; 125 A Crim R 315; \[2001\] NSWCCA 413](#) at [60], [98]–[100]. See also [R v JCG \(2001\) 127 A Crim R 493; \[2001\] NSWCCA 504](#).
- 158 Hon FHR Vincent, *Report – Inquiry into the circumstances that led to the conviction of Mr Farah Abdulkadir Jama* (Victorian Government Printer, May 2010), p 45. See also [Forbes v The Queen \(2009\) 167 ACTR 1; 232 FLR 229; \[2009\] ACTCA 10](#) at [19]–[41] where the court concluded the contention DNA evidence alone could not establish guilt beyond reasonable doubt was not supported by any rule or principle.
- 159 [R v Green](#) (unreported, NSWCA, 26 March 1993); see also [R v Pantoja \(1996\) 88 A Crim R 554](#) at 564, per Hunt CJ at CL; at 584–584, per Abadee J, Hidden J agreeing with Hunt CJ at CL.
- 160 Hon FHR Vincent, *Report – Inquiry into the circumstances that led to the conviction of Mr Farah Abdulkadir Jama* (Victorian Government Printer, May 2010), p 38.
- 161 [Forbes v The Queen \[2010\] HCATrans 120](#), special leave refused on appeal from [Forbes v The Queen \(2009\) 167 ACTR 1; 232 FLR 229; \[2009\] ACTCA 10](#).
- 162 Hon FHR Vincent, *Report – Inquiry into the circumstances that led to the conviction of Mr Farah Abdulkadir Jama* (Victorian Government Printer, May 2010), pp 37, 39. The defence called no evidence in support of this contention.
- 163 [Forbes v The Queen \(2009\) 167 ACTR 1; 232 FLR 229; \[2009\] ACTCA 10](#).
- 164 [Forbes v The Queen \(2009\) 167 ACTR 1; 232 FLR 229; \[2009\] ACTCA 10](#) at [33]–[36].
- 165 This dilemma was raised in the special leave application before the High Court in [Forbes v The Queen \[2010\] HCATrans 120](#), see lines 2526–2555, but it was not explored in any depth. Special leave was refused by the court on the basis it was open to the jury to conclude from the evidence led in the trial that the accused was guilty beyond reasonable doubt and given the way the parties conducted the trial it was not a suitable case (at lines 2752–2755) to consider the broader question the applicant sought to agitate, namely the conviction was unsafe because the only evidence against him was the DNA evidence (at lines 15–21).
- 166 See brief discussion of Locard at [\[ID.3160\]](#) “Trace DNA – Low copy number typing”.
- 167 [Fitzgerald v The Queen \[2014\] HCA 28](#). It would appear the only prosecution evidence linking the accused [Fitzgerald \[2014\] HCA 28](#) to the offence was the uncontested evidence his DNA was found in one sample from the didgeridoo. The DNA of the deceased and another injured occupant of the house were also found in apparent bloodstains on the didgeridoo. Fitzgerald did not give evidence on his trial. However it would appear his co-accused Sumner gave evidence he had shaken hands with Fitzgerald at least twice at an unrelated location before

he attended the house where the attack took place some hours before the fatal assault. Whilst at that house Sumner had got into an altercation, suffered a fractured jaw and it was alleged threatened retaliation as he left. It would appear from the High Court's reasons Sumner admitted he attended the house a second time, at the time of the fatal attack, but denied inflicting the fatal injuries. See also *Director of Public Prosecutions (Vic) v Wise* [2016] VSCA 173 at [65]–[67] per curiam, where *Fitzgerald* [2014] HCA 28 is considered.

- 168 *R v Guingab* [2010] VSC 256 at [31], per Forrest J. The evidence may be
excluded under *Evidence Act 2008*, s 135 and must be excluded if s 137 applies.
- 169 Hon FHR Vincent, *Report – Inquiry into the circumstances that led to the
conviction of Mr Farah Abdulkadir Jama* (Victorian Government Printer, May
2010), p 43.
- 170 *R v Watters* [2000] EWCA Crim 89 where on appeal held a no case submission
should have been upheld at trial. However, caution should be applied to the
decision in *R v Watters* as the appeal court was careful to confine their remarks to
the case before them, at [21]. The forensic DNA expert called by the prosecution
conceded in cross examination that if the accused had two brothers the match
probabilities would be reduced substantially. The evidence showed the accused
had two brothers who had not provided reference samples and that one of the two
brothers was suspected of being a member of the safe cracking team. See [17]–
[20], [22]. Compare *R v Tsekiri* [2017] EWCA Crim 40 where the accused's DNA
was found on the driver's door handle used to gain entry to the vehicle and rob
the driver. Held even though the DNA was the only evidence against the accused
and could have been transferred to the door handle at any number of locations,
given the vehicle was a movable object, there was a case to answer. The match
probability of 1:1 billion was sufficient, without more, to raise a case to answer.
- 171 *Fitzgerald v The Queen* [2014] HCA 28.
- 172 *R v Adams (No 1)* [1996] 2 Cr App R 467 at 469–470. By the phrase “None of
these provides” Rose LJ referred to a number of cases discussed earlier in the
reasons. See *R v FNC* [2015] EWCA Crim 1732 for a review of relevant English
cases, including *Doheny v The Queen* [1997] 1 Cr App R 369 and *R v Sampson
and Kelly* [2014] EWCA Crim 1968, on the question whether an accused can be
convicted on DNA evidence alone. See also *R v Tsekiri* [2017] EWCA Crim 40.
- 173 *Jury Directions Act 2015*, s 35.
- 174 *Jury Directions Act 2015*, s 3.
- 175 *Jury Directions Act 2015*, s 61(a).
- 176 *R v Green* (unreported, NSWCA, 26 March 1993); see also *R v Pantoja* (1996)
88 A Crim R 554, per Abadee J; *R v Milat* (1996) 87 A Crim R 446 at 447, per
Hunt CJ at CL.
- 177 *R v Pantoja* (1996) 88 A Crim R 554 at 560.

- 178 *R v Pantoja* (1996) 88 A Crim R 554 at 564.
- 179 *R v Berry*; *R v Wenitong* (2007) 17 VR 153; 176 A Crim R 195; [2007] VSCA 202 at [57], per Redlich JA, Buchanan JA agreeing and Kellam JA agreeing in a separate judgment at [141].
- 180 *R v Berry*; *R v Wenitong* (2007) 17 VR 153; 176 A Crim R 195; [2007] VSCA 202 at [59], per Redlich JA, Buchanan JA agreeing and Kellam JA agreeing in a separate judgment at [141].
- 181 *Latcha v The Queen* (1998) 8 NTLR 122; 148 FLR 365; 104 A Crim R 390; 127 NTR 1 at 396–397 (A Crim R).
- 182 *Doheny v The Queen* [1997] 1 Cr App R 369.
- 183 *R v Karger* (2001) 83 SASR 1; [2001] SASC 64, per Mullighan J; *R v McIntyre* [2001] NSWSC 311, per Bell J; *R v Gallagher* [2001] NSWSC 462, per Barr J. In *R v Gallagher* the admissibility of Profiler Plus results was challenged on the basis they were not reliable because the New South Wales Department of Analytical Laboratories (DAL) did not have access to the primer sequences (used in the PCR amplification process) and could not therefore validate the system and thus the results. Barr J, after examining the system of accreditation required of Australian forensic laboratories and the requirements for ongoing assessment of the laboratories, as well as reviews and assessments of Profiler Plus in the scientific literature, found that Profiler Plus had been satisfactorily validated in DAL before the subject tests were carried out.
- 184 *Crimes Act 1958*, s 464(2).
- 185 *Crimes Act 1958*, s 464(2).
- 186 *Crimes Act 1958*, s 464(2).
- 187 *Crimes Act 1958*, s 464(2).
- 188 *Crimes Act 1958*, s 464(2).
- 189 *Crimes Act 1958*, s 464(2).
- 190 *Crimes Act 1958*, s 464(2).
- 191 *Crimes Act 1958*, s 464R(1)(a)–(c).
- 192 *Crimes Act 1958*, s 464R(2)(a)–(b).
- 193 *Crimes Act 1958*, s 464S(1)(a)–(g).
- 194 *Martin v R (No 2)* [2022] VSCA 161 at [79] per curiam.
- 195 *Martin v R (No 2)* [2022] VSCA 161 at [86] – [100] per curiam.
- 196 *Crimes Act 1958*, s 464S(2).
- 197 *R v Su* (2003) 7 VR 13; 142 A Crim R 338; [2003] VSC 305 at [72], per Coldrey J.
- 198 A “relevant suspect” is defined in *Crimes Act 1958*, s 464(2) as a person of or above the age of 18 who is suspected of having committed or having attempted to commit or who has been charged with having committed an indictable offence against the person at common law or an indictable offence under Div 1 of Pt I or under ss 75, 75A, 76, 77, 197 (where the offence is charged as arson), 197A,

- 248 or 318 or under *Drugs, Poisons and Controlled Substances Act 1981*, ss 71, 72(1)(ab) or (b).
- 199 *Crimes Act 1958*, s 464T(1)(a)–(d).
- 200 *Crimes Act 1958*, s 464T(3)(a).
- 201 *Crimes Act 1958*, s 464T(3)(b).
- 202 *Fawcett v Nimmo* (2005) 156 A Crim R 431; [2005] NSWSC 1047 at [14], per Grove J. Whilst Grove J was referring to the *Crimes (Forensic Procedures) Act 2000* (NSW) it might be argued his comments are apposite to the equivalent powers contained in the *Crimes Act 1958*. See also *Orban v Bayliss* [2004] NSWSC 428 at [30].
- 203 *Orban v Bayliss* [2004] NSWSC 428 at [40]–[48].
- 204 *Fawcett v Nimmo* (2005) 156 A Crim R 431; [2005] NSWSC 1047 at [15], per Grove J, but note it was a specific requirement of the New South Wales Act being considered that the Magistrate be satisfied “on the evidence before the court” and this phrase does not appear in the *Crimes Act 1958* provision.
- 205 *Maguire v Beaton* (2005) 162 A Crim R 21; [2005] NSWSC 1241.
- 206 *Crimes Act 1958*, s 464T(3)(c)(i)–(iv).
- 207 *Crimes Act 1958*, s 464T(3)(d)–(g).
- 208 *Crimes Act 1958*, s 464T(3)(h).
- 209 *Crimes Act 1958*, s 464T(5) and note *Charter of Human Rights and Responsibilities Act 2006*, ss 8, 24 and 25 (the Charter) are potentially in conflict with the limitations imposed on the suspect's ability to participate in the application to the Magistrates' Court. An issue may be whether such an application is a civil proceeding, an administrative function or has been made after the person has been charged with a criminal offence. King J in *R v Williams* (2007) 16 VR 168; [2007] VSC 2 at [52] took the view that Parliament's intention was for the courts to become actively involved in the interpretation of the Charter and human rights only after 1 January 2008.
- 210 *Crimes Act 1958*, s 464SA.
- 211 See *Crimes Act 1958*, s 464T(3)(c)–(f).
- 212 *Crimes Act 1958*, s 464SA(2), (6).
- 213 *Crimes Act 1958*, s 464SA(3), (5).
- 214 *Crimes Act 1958*, s 464SA(4).
- 215 *Crimes Act 1958*, s 464SB(2).
- 216 See, for example, *Stevenson v Ellis* (1996) 85 A Crim R 49; *Ellis v Stevenson* (1996) 86 A Crim R 368.
- 217 See, for example, *Stevenson v Ellis* (1996) 85 A Crim R 49; *Ellis v Stevenson* (1996) 86 A Crim R 368.
- 218 *Crimes Act 1958*, s 464U(1)(a)–(b).
- 219 *Crimes Act 1958*, s 464U(2).

- 220 Including offences against *Crimes Act 1958*, Div 1 of Pt I, under ss 75, 75A, 76, 77, 197 (where it is arson), ss 197A, 249, 250, 251, 317, 317A, 318; *Drugs, Poisons and Controlled Substances Act 1981*, ss 71–71AA, 72, 72A.
- 221 *Crimes Act 1958*, s 464U(7)(a).
- 222 *Crimes Act 1958*, s 464U(7)(b).
- 223 *Crimes Act 1958*, s 464U(7)(c).
- 224 *Crimes Act 1958*, s 464U(7)(d)–(f).
- 225 *Crimes Act 1958*, s 464U(7)(g).
- 226 *Crimes Act 1958*, s 464U(8)(a)–(c).
- 227 *Crimes Act 1958*, s 464U(12) and note *Charter of Human Rights and Responsibilities Act 2006*, ss 8, 24 and 25 (the Charter) are potentially in conflict with the limitations imposed on the suspect's ability to participate in the application to the Children's Court. An issue may be whether such an application is a civil proceeding, an administrative function or has been made after the person has been charged with a criminal offence. King J in *R v Williams* (2007) 16 VR 168; [2007] VSC 2 at [52] took the view that Parliament's intention was for the courts to become actively involved in the interpretation of the Charter and human rights only after 1 January 2008.
- 228 *Crimes Act 1958*, s 464V(1).
- 229 *Crimes Act 1958*, s 464V(1).
- 230 *Crimes Act 1958*, s 464V(4).
- 231 *Crimes Act 1958*, s 464X(1)(a)–(f). The power does not extend to the granting of warrants in respect of “detained” and “protected persons”.
- 232 *Crimes Act 1958*, s 464Y(1), (2).
- 233 *Crimes Act 1958*, s 464Z(3)(a).
- 234 *Crimes Act 1958*, s 464Z(3)(b).
- 235 *Crimes Act 1958*, s 464Z(3)(c). As to who is authorised see *Crimes Act 1958*, s 464Z(1). The Chief Commissioner has, for the purpose of supervising a self-taken mouth scraping, authorised any police member who has completed the *Crimes (DNA Database) Act 2002* training – see *Victoria Police Manual*, section 3 VPM 111–7 Aids to Investigation – Forensic Procedures.
- 236 *Crimes Act 1958*, s 464Z(3A) and (3B).
- 237 *Crimes Act 1958*, s 464Z(3AA), (3AB), (3B).
- 238 *Crimes Act 1958*, s 464Z(4).
- 239 *Crimes Act 1958*, s 464Z(6)(a).
- 240 *Crimes Act 1958*, s 464Z(6)(ab), (9).
- 241 *Crimes Act 1958*, s 464Z(6)(b), (c).
- 242 *Crimes Act 1958*, s 464Z(7).
- 243 *Crimes Act 1958*, s 464ZA(1) and (2).
- 244 *Crimes Act 1958*, s 464ZA(4)(a)–(b).

- 245 In particular see *Victoria Police Manual*, VPM 114–4 Exhibits and seized
property and sections 7 and 8 VPM 111–7 Aids to investigation – Forensic
procedures.
- 246 *Crimes Act 1958*, s 464ZB(1)(a)–(b).
- 247 *Crimes Act 1958*, s 464ZC(1)–(2).
- 248 *Crimes Act 1958*, s 464ZC(3)–(4).
- 249 *Crimes Act 1958*, s 464ZD.
- 250 Pursuant to *Crimes Act 1958*, ss 464ZGB–464GD.
- 251 *Crimes Act 1958*, s 464ZE(2)(a)–(b).
- 252 *Crimes Act 1958*, s 464ZE(2)(a)–(h).
- 253 *Crimes Act 1958*, s 464ZE(3).
- 254 *Drugs, Poisons and Controlled Substances Act 1981*, s 71.
- 255 *Drugs, Poisons and Controlled Substances Act 1981*, s 72(1)(ab).
- 256 *Drugs, Poisons and Controlled Substances Act 1981*, s 72(1)(b).
- 257 *Crimes Act 1958*, s 464ZF(2).
- 258 *Crimes Act 1958*, s 464ZFA(1).
- 259 *Crimes Act 1958*, s 464ZF(3). See eg *Lednar v Magistrates' Court* (2000) 117
A Crim R 396; [2000] VSC 549; *Pavic v Magistrates Court (Vic)* (2003) 140 A
Crim R 113; [2003] VSC 99.
- 260 *Crimes Act 1958*, s 464ZF(4).
- 261 *Crimes Act 1958*, s 464ZF(8)(a)–(c).
- 262 *R v Abebe* [1999] VSC 214, per Harper J.
- 263 *Lednar v Magistrates' Court* (2000) 117 A Crim R 396; [2000] VSC 549 at [242]–
[244].
- 264 *Lednar v Magistrates' Court* (2000) 117 A Crim R 396; [2000] VSC 549 at [287].
- 265 *Lednar v Magistrates' Court* (2000) 117 A Crim R 396; [2000] VSC 549 at [301].
- 266 *Lednar v Magistrates' Court* (2000) 117 A Crim R 396; [2000] VSC 549 at [361].
- 267 *Lednar v Magistrates' Court* (2000) 117 A Crim R 396; [2000] VSC 549 at [380].
- 268 *Lednar v Magistrates' Court* (2000) 117 A Crim R 396; [2000] VSC 549 at [449].
- 269 *Lednar v Magistrates' Court* (2000) 117 A Crim R 396; [2000] VSC 549 at [453].
- 270 *Crimes Act 1958*, s 464ZFB(1).
- 271 *MB v Children's Court of Victoria & Anor* [2023] VSC 666 at [15] – [25] per
McDonald J.
- 272 Pursuant to *Crimes Act 1958*, s 464ZFB .
- 273 *Crimes Act 1958*, s 464ZFC(1).
- 274 *Crimes Act 1958*, s 464ZFD(1).
- 275 *Crimes Act 1958*, s 464ZFD(2).
- 276 *Crimes Act 1958*, s 464ZFE.
- 277 *Crimes Act 1958*, s 464ZFG(2).
- 278 *Crimes Act 1958*, s 464ZFG(1).

279	<i>Crimes Act 1958</i> , s 464ZFG(5).
280	<i>Crimes Act 1958</i> , s 464ZFG(6).
281	<i>Crimes Act 1958</i> , s 464ZGA(1).
282	<i>Crimes Act 1958</i> , s 464ZGB(3).
283	<i>Crimes Act 1958</i> , s 464ZGE(1).
284	<i>Crimes Act 1958</i> , s 464ZGE(2).
285	<i>Crimes Act 1958</i> , s 464ZGF(2)(a)–(b).
286	<i>Crimes Act 1958</i> , s 464ZF(2)(a)–(b).
287	<i>Crimes Act 1958</i> , s 464ZF(3)(a)–(b).
288	<i>Crimes Act 1958</i> , s 464ZF(8).
289	<i>Crimes Act 1958</i> , s 464ZFA(1).